# Intro to GIS Exercise: QGIS

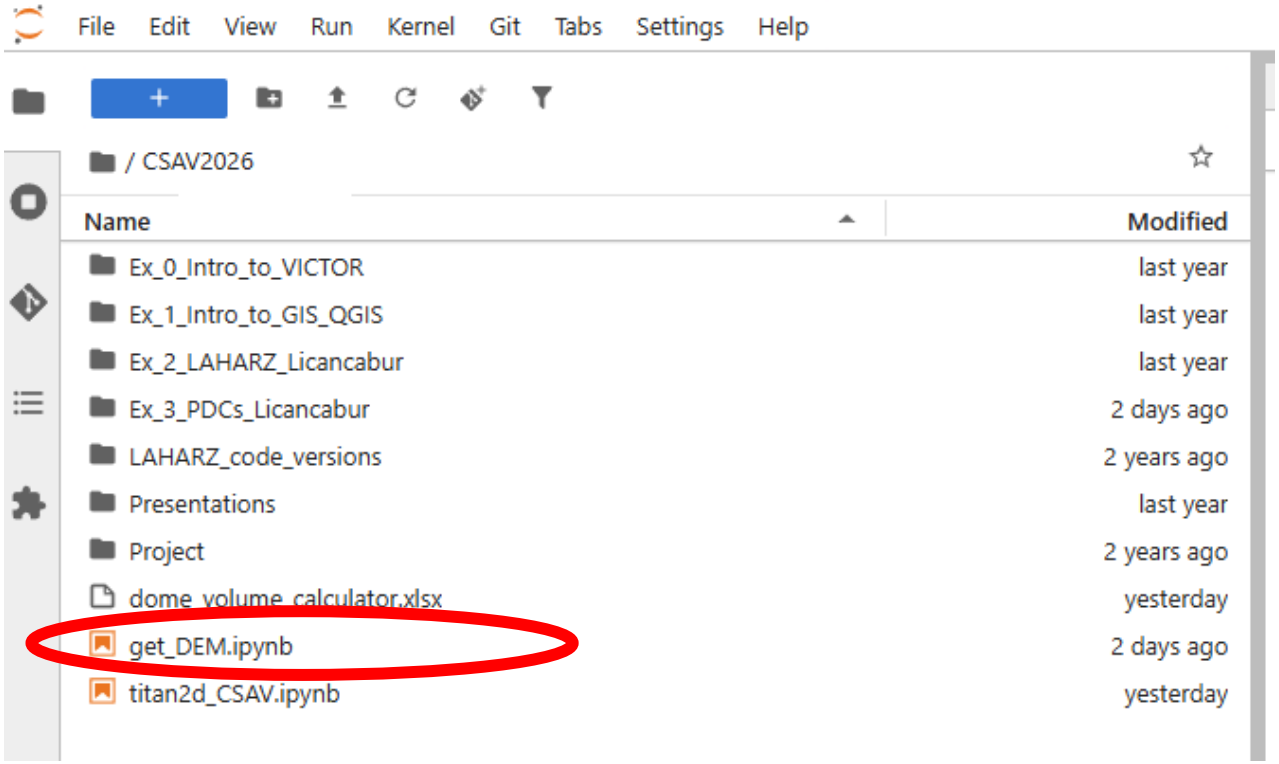


# Exercise Goals

- **Learn to use VICTOR to download a DEM**
- **Get comfortable with QGIS basics**

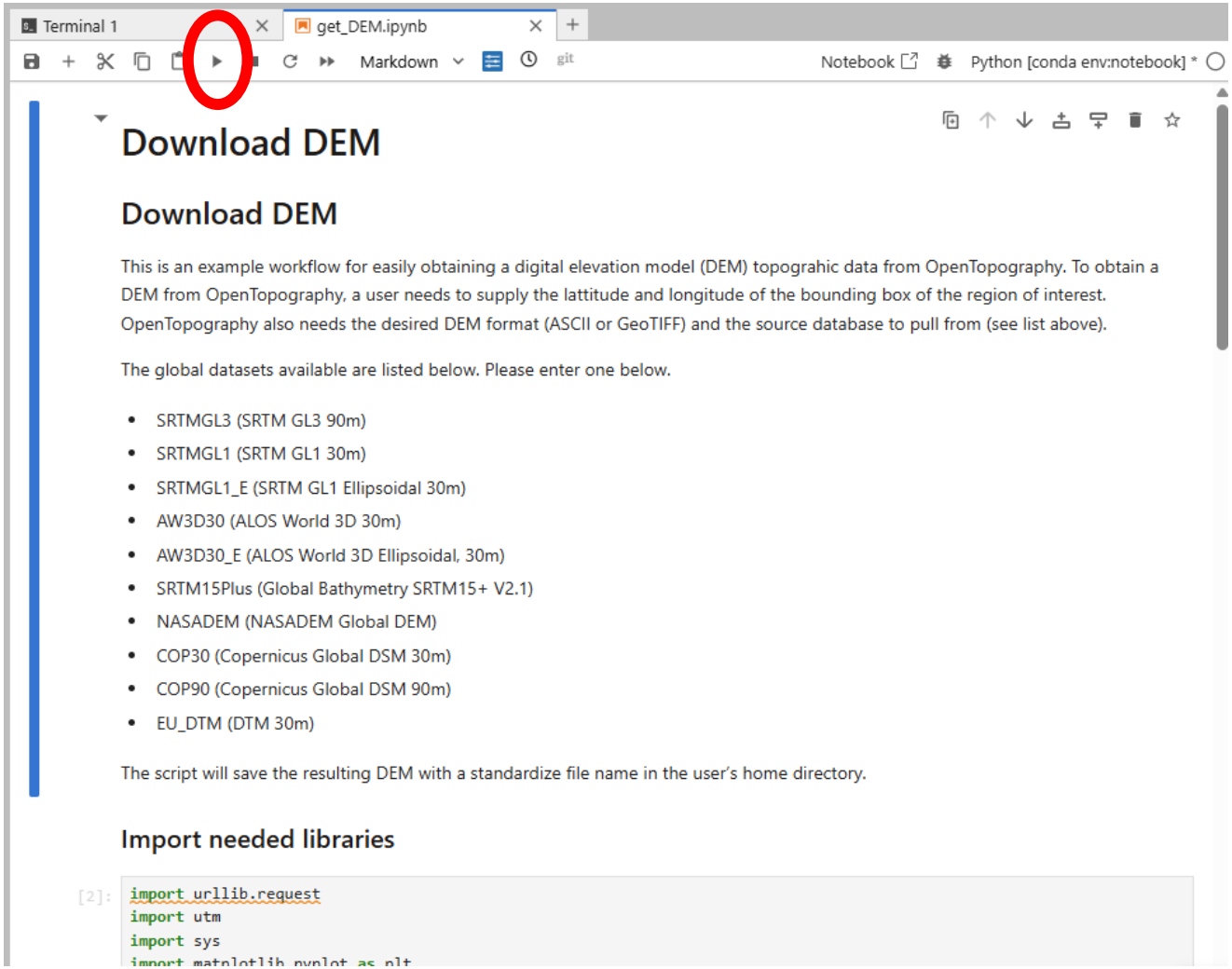
# VICTOR DEM exercise

# Using VICTOR to get a DEM




- Inside VICTOR, you should have a Jupiter notebook called “get_DEM.ipynb”
- Open it

- There are some instructions in the Jupyter notebook itself, but our specific instructions to download the correct DEM are in this presentation
- To run the cells, put your cursor in the first box and click the run button at the top
- You can click through the cells using the run button



## Import needed libraries

```
#Don't change anything here, just run
import urllib.request
import utm
import sys
import matplotlib.pyplot as plt
import cartopy.crs as ccrs
import rasterio as rio
import xarray as xr

import openpyxl
```

[3]:
```
#Don't change anything here, just run
sys.path.insert(0,"/home/jovyan/shared/Libraries")
import victor
```

- For some boxes, you don't change any parameters, you just run it!
- When you run a cell, the box next to it will display [*] while it is running and will change to [2] (or another number) when it is done

## Setting input parameters

In the following cell we use the lat/lon of a target volcano's summit, obtained from the Global Volcano Project database from the volcano'es name, to set the bounding box.

[3]:
```
#Choose a volcano and DEM options

volcano = victor.search_volcano('St. Helens') # Get volcano information from the GVP database

center_lat = volcano[1][0]  # Get volcano latitude
center_lon = volcano[2][0]  # Get volcano longitude

# Set the extent of the region of interest, in meters in each direction from the summit
eastwest_extent = 30000
northsouth_extent = 30000

# Set bounding box in lat/lon
utms = utm.from_latlon(center_lat, center_lon)
[north_lat, east_lon] = utm.to_latlon(utms[0]+eastwest_extent, utms[1]+northsouth_extent, utms[2], utms[3])
[south_lat, west_lon] = utm.to_latlon(utms[0]-eastwest_extent, utms[1]-northsouth_extent, utms[2], utms[3])

# Select a DEM format: Input "ascii" for ASCII format (.asc) and "tiff" for geotiff format
outputFormat = "tiff"

# Enter the name of the output file (without extension)
name = "St. Helens"

dataset = "SRTMGL1"

print(center_lat, center_lon)
```

- This code uses the GVP database to get the summit coordinates of a volcano you enter.
- Change the volcano to 'St. Helens'
- Troubleshooting: this needs to match the GVP name exactly!
- Make sure the parameters look like the ones in this screenshot. If they don't, change the parameters
- When it is done running, it will print the center coordinates under this box, which should look like this:

```
print(center_lat, center_lon)
```

```
46.2 -122.18
```

## Obtain DEM

This cells obtains the DEM from OpenTopograhy using the input options. It will print out the resulting file name.

```
[*]: dem = victor.download_dem(north_lat, south_lat, east_lon, west_lon, outputFormat, dataset)
print (dem)
```

- When you run the “Obtain DEM” code, you will see that it takes some time to run the code.
- You can tell because there is a [*] in the brackets
- When the * changes to a number, you should see a file downloaded in your home folder!

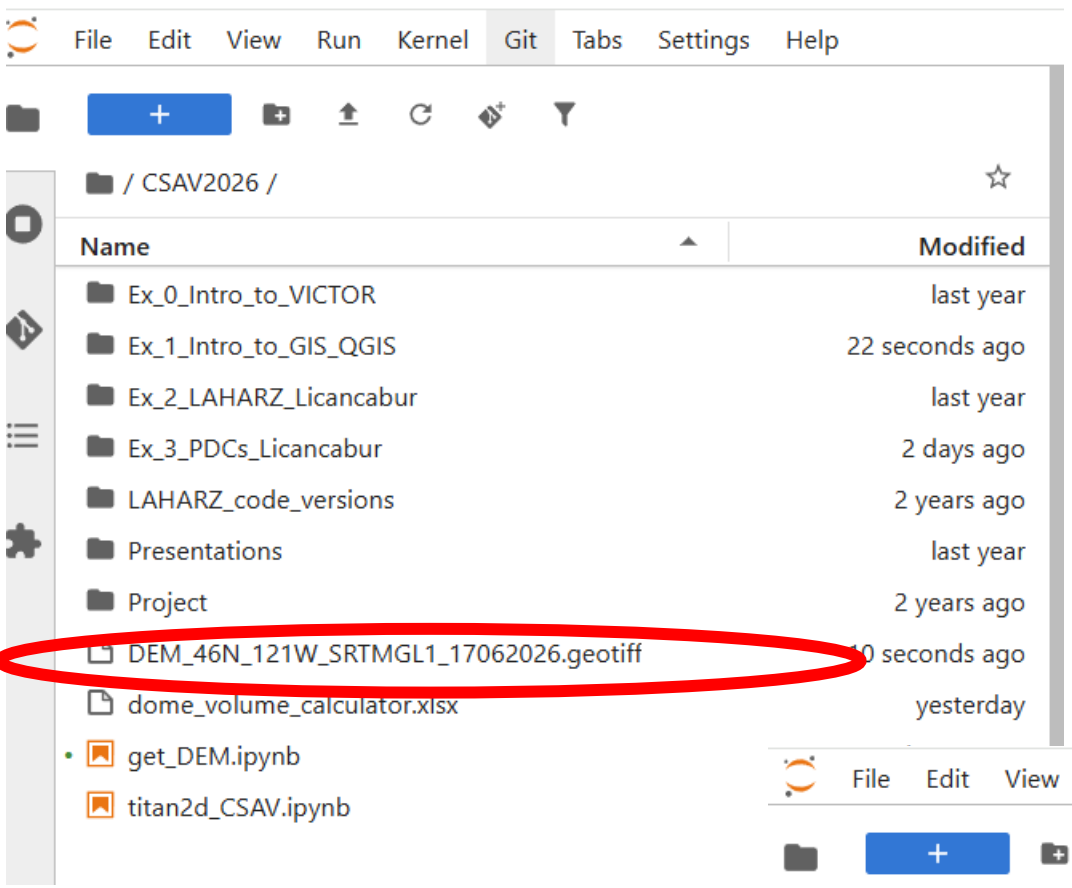
## Obtain DEM

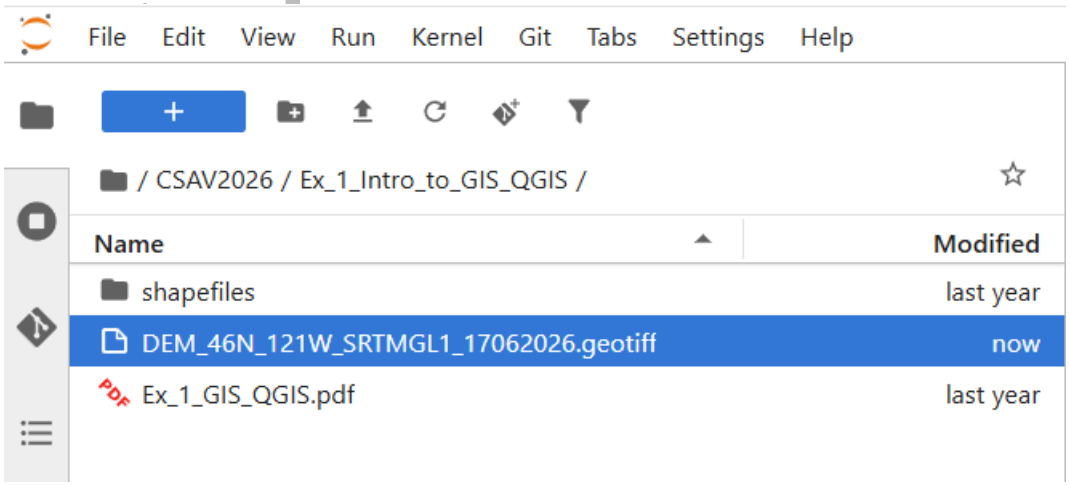
This cells obtains the DEM from OpenTopograhy using the input options. It will print out the resulting file name.

```
[11]: dem = victor.download_dem(north_lat, south_lat, east_lon, west_lon, outputFormat, dataset)
print (dem)
```

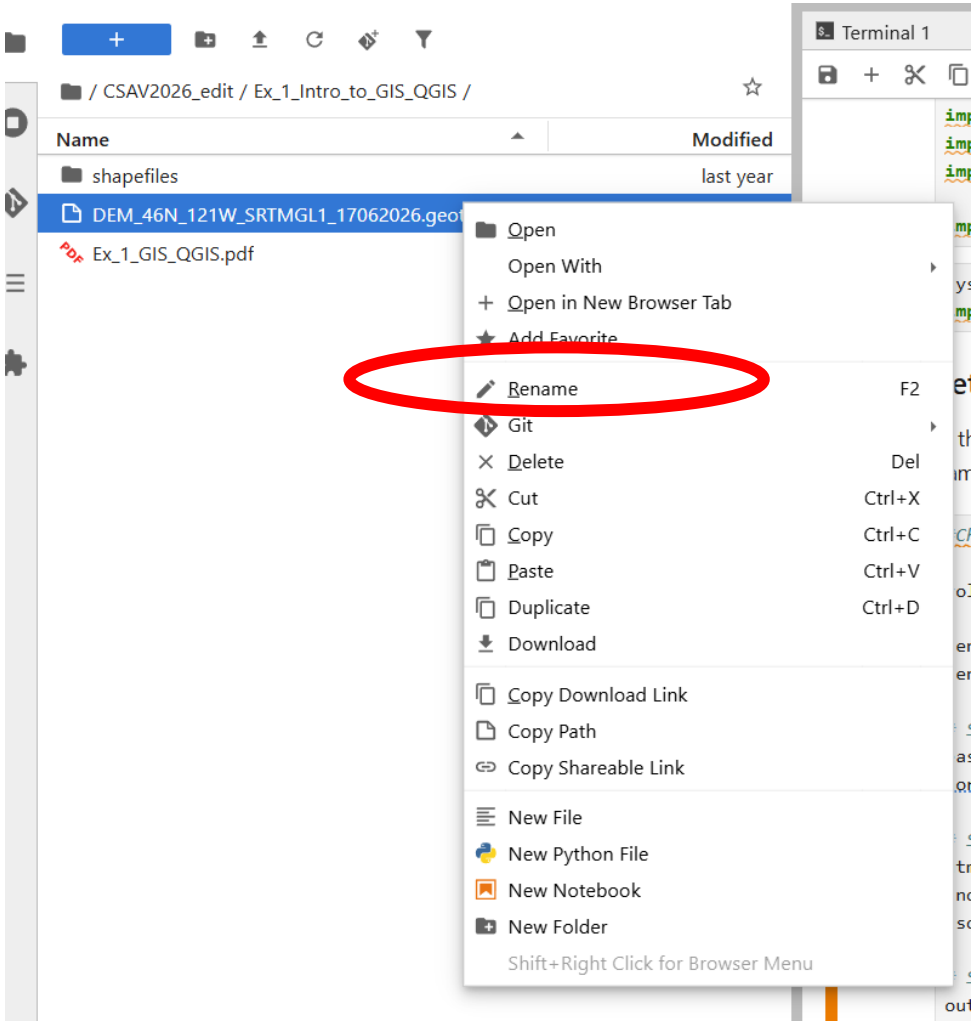
```
DEM downloaded as DEM_46N_121W_SRTMGL1_17062026.geotiff
DEM_46N_121W_SRTMGL1_17062026.geotiff
```

- Note: the next cell just prints an image of the DEM. It will fail because it doesn’t like the space in St. Helens



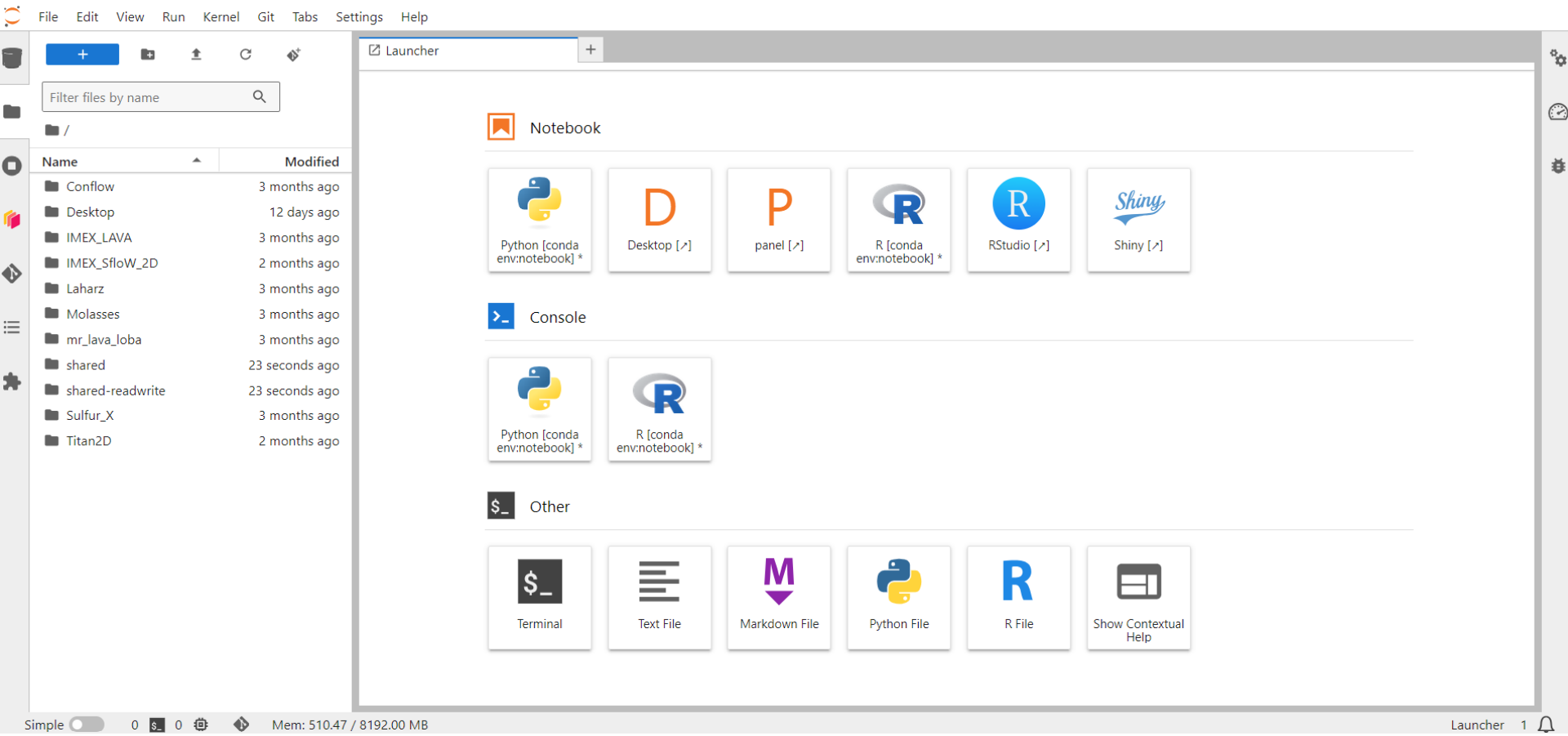




- Copy the file to the Ex_1_Intro_to_GIS_QGIS folder



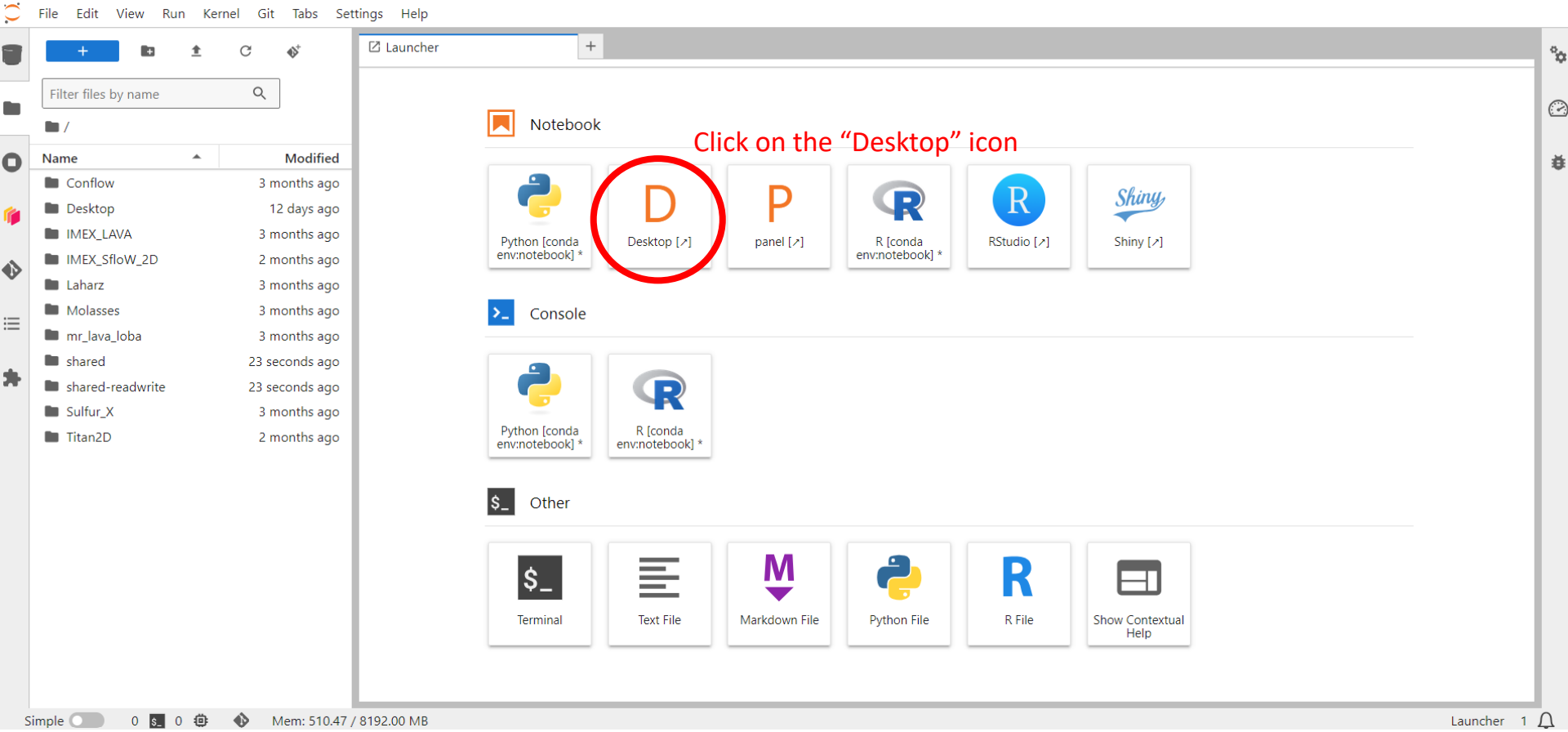

- Rename the file to “msh_gcs.tif”
- GCS=geographic coordinate system

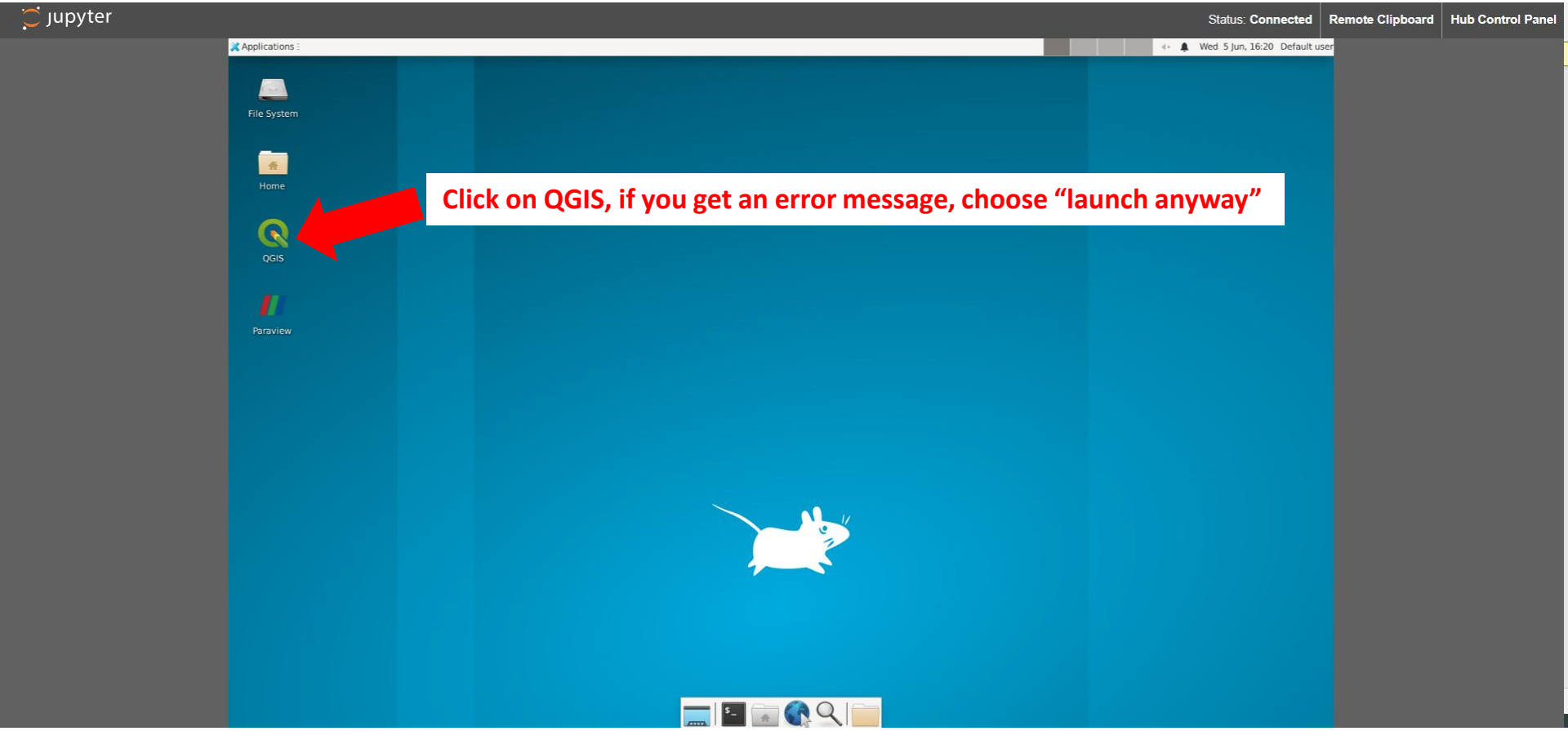
# QGIS exercise

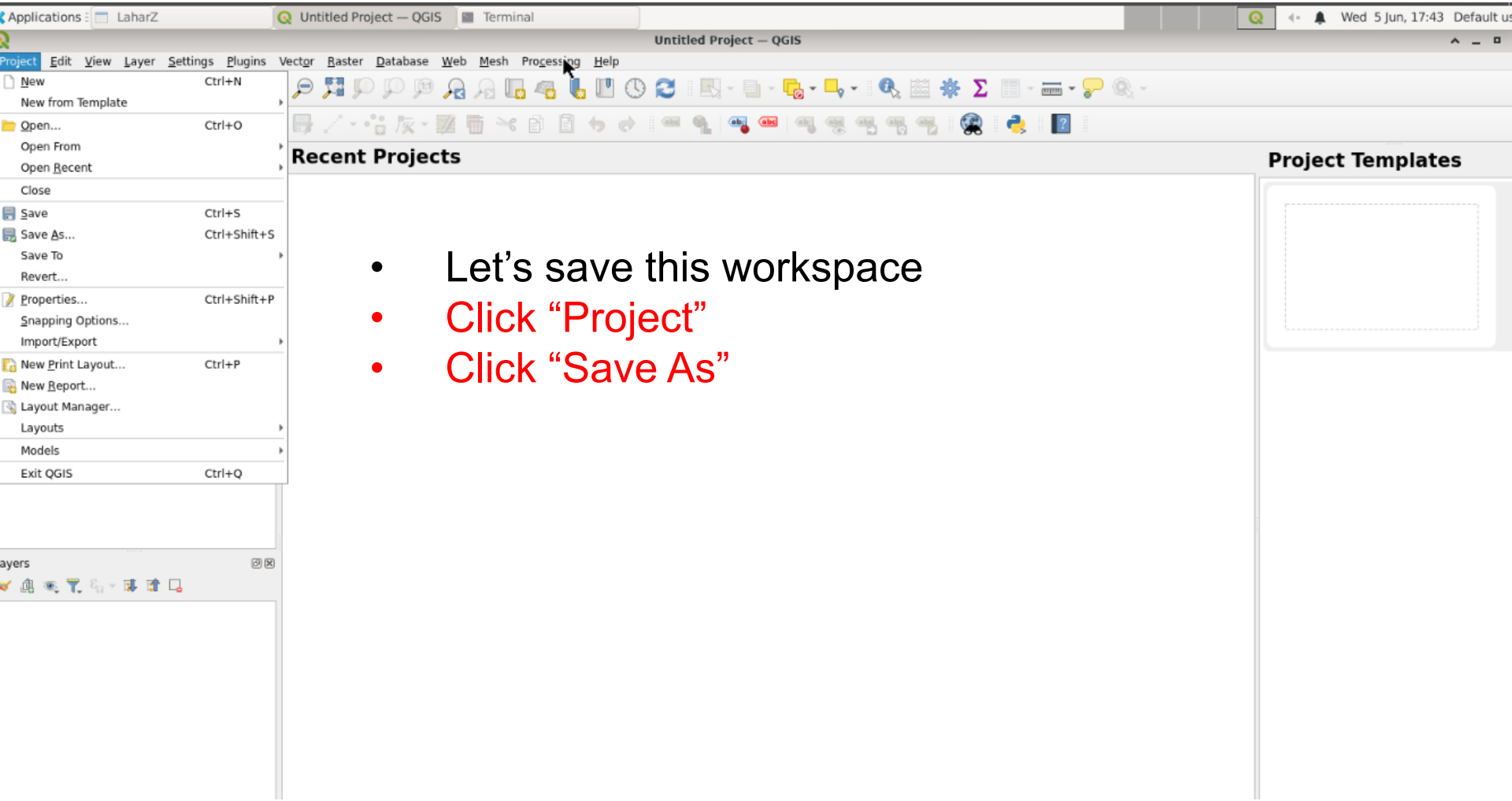



- When you login to VICTOR, your screen will look something like this
- Your files are on the left panel
- Your “Launcher” has useful icons

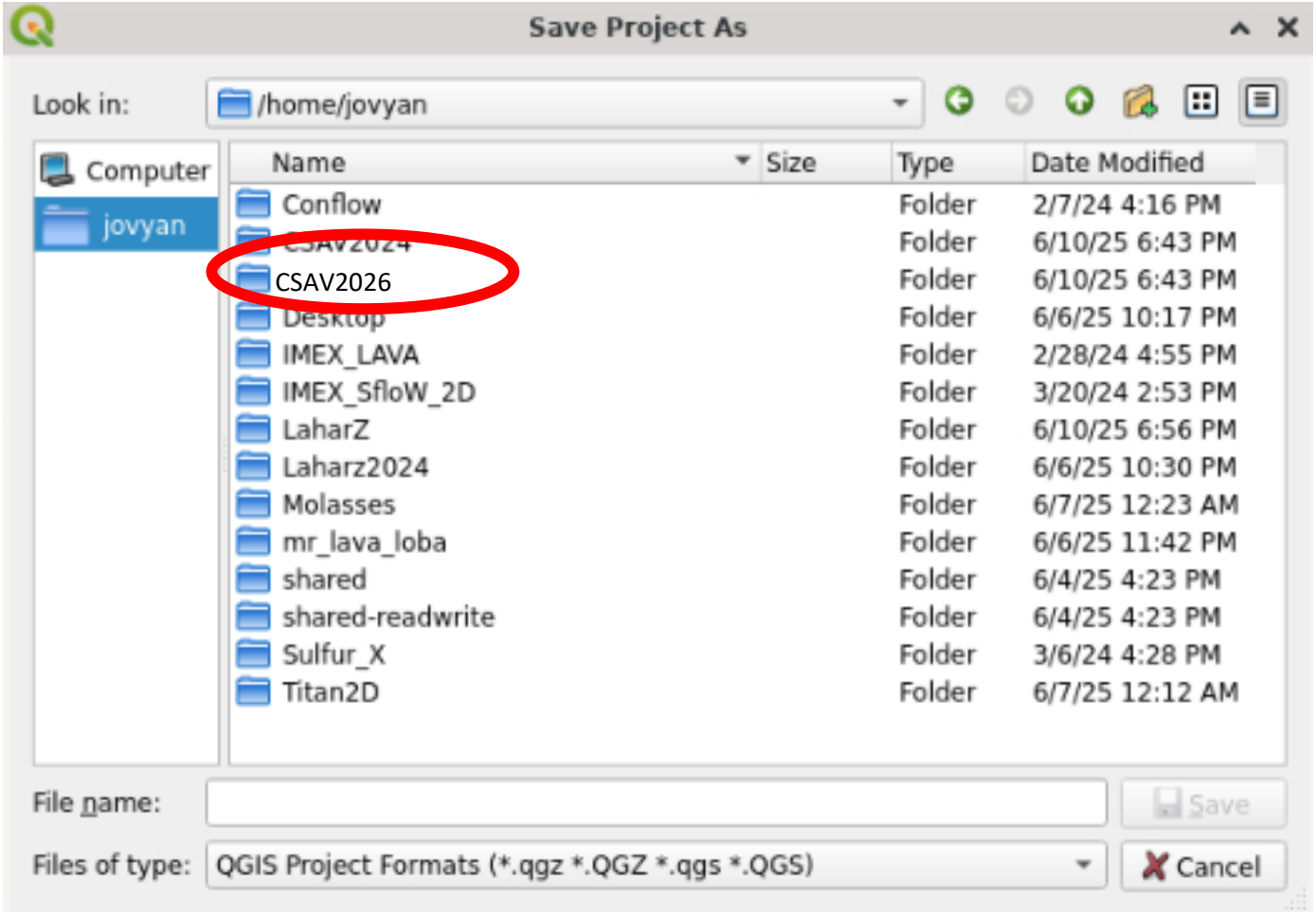
Tip: If you lose your “Launcher” screen, click the “plus” icon at the top left for “New Launcher”


File Edit View Run Kernel Git Tabs Settings Help
Launcher
Filter files by name
/
Name
Modified
Conflow 3 months ago
Desktop 12 days ago
IMEX_LAVA 3 months ago
IMEX_SfloW_2D 2 months ago
Laharz 3 months ago
Molasses 3 months ago
mr_lava_loba 3 months ago
shared 23 seconds ago
shared-readwrite 23 seconds ago
Sulfur_X 3 months ago
Titan2D 2 months ago
Notebook
Click on the “Desktop” icon
Python [conda env:notebook] *
Desktop [↗]
panel [↗]
R [conda env:notebook] *
RStudio [↗]
Shiny [↗]
Console
Python [conda env:notebook] *
R [conda env:notebook] *
Other
Terminal
Text File
Markdown File
Python File
R File
Show Contextual Help
Simple
0 0
Mem: 510.47 / 8192.00 MB
Launcher 1

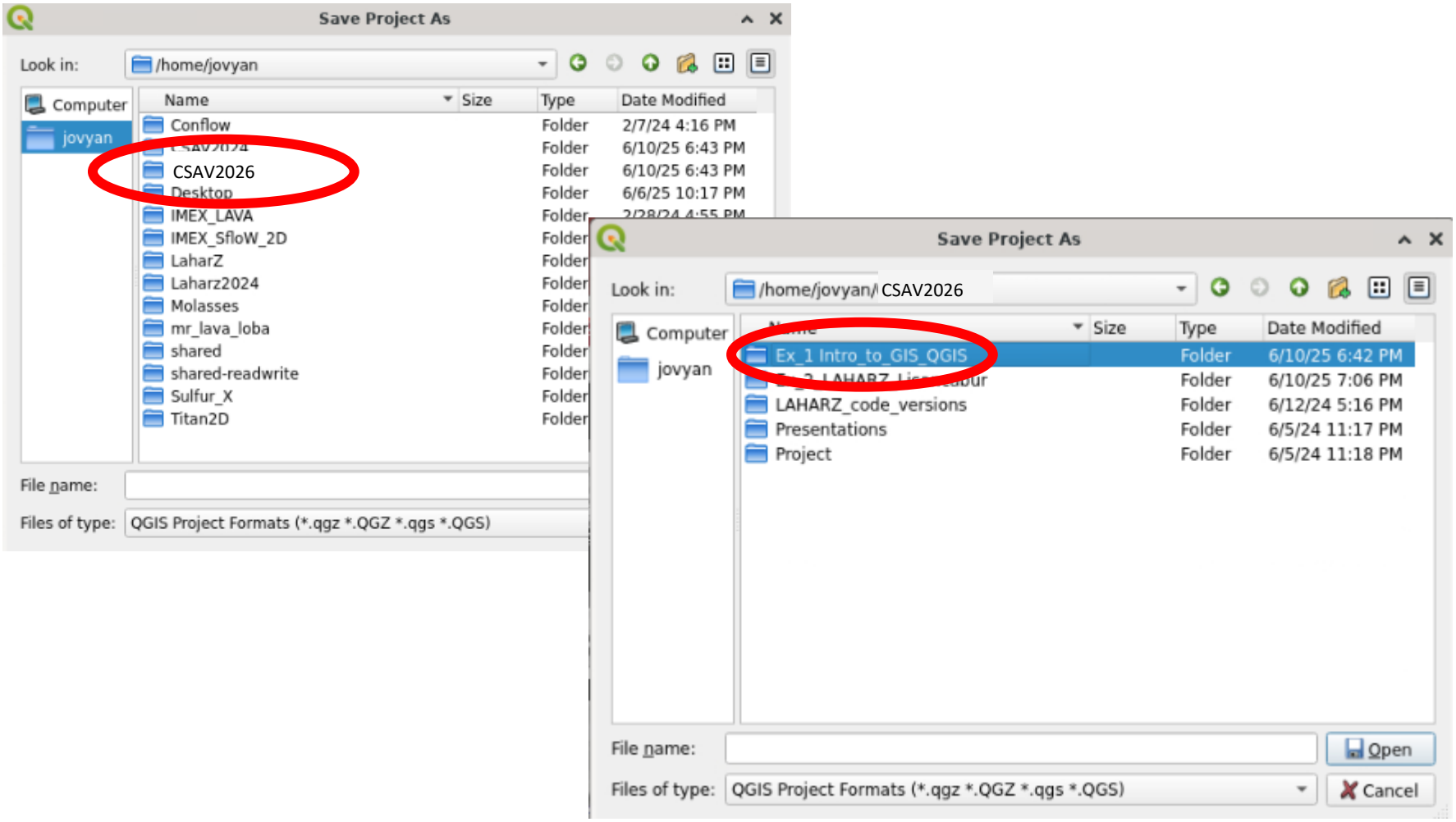

jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Wed 5 Jun, 16:20 Default user
File System
Home
QGIS
Paraview
Click on QGIS, if you get an error message, choose “launch anyway”



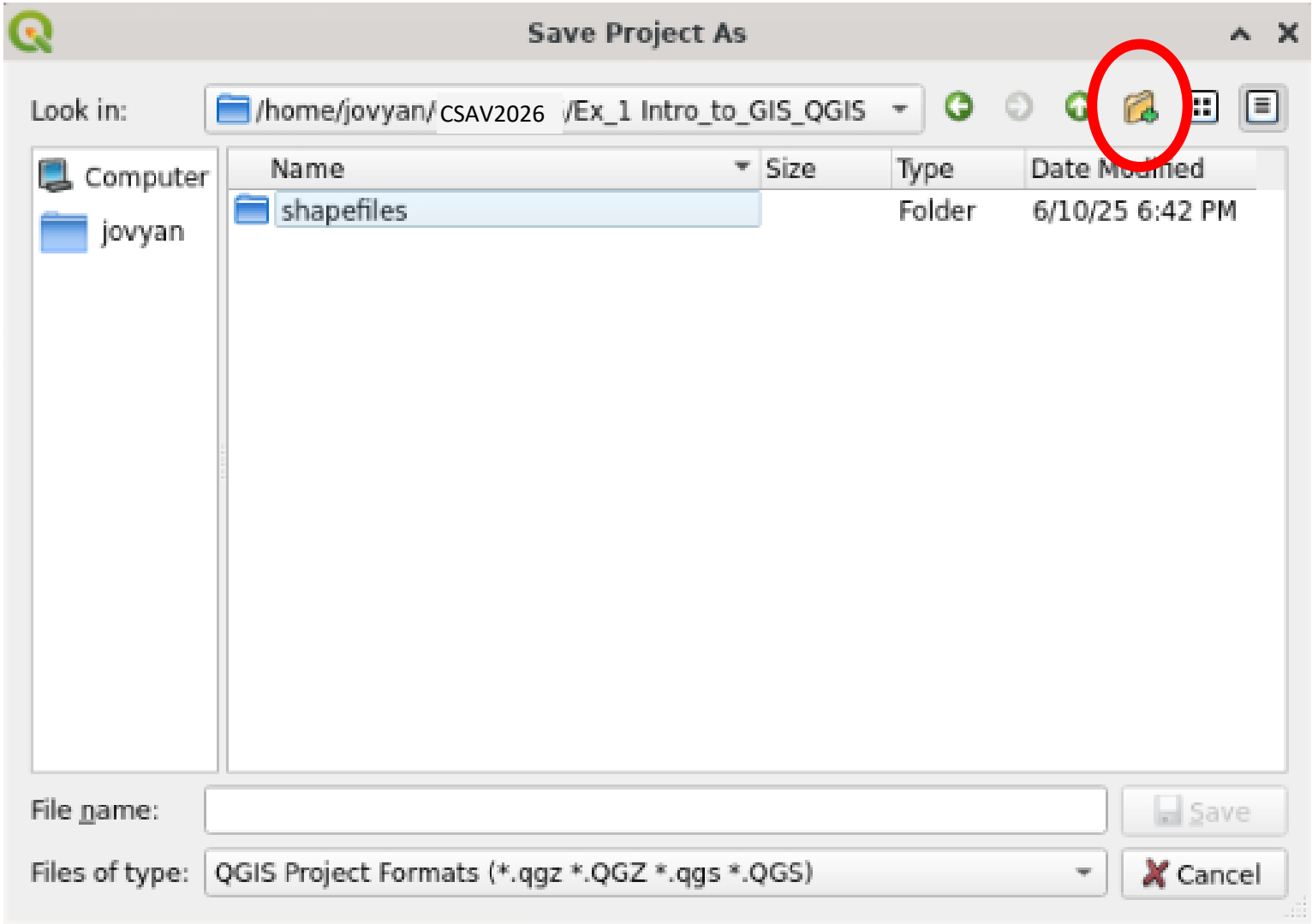

- Let’s save this workspace
- Click “Project”
- Click “Save As”



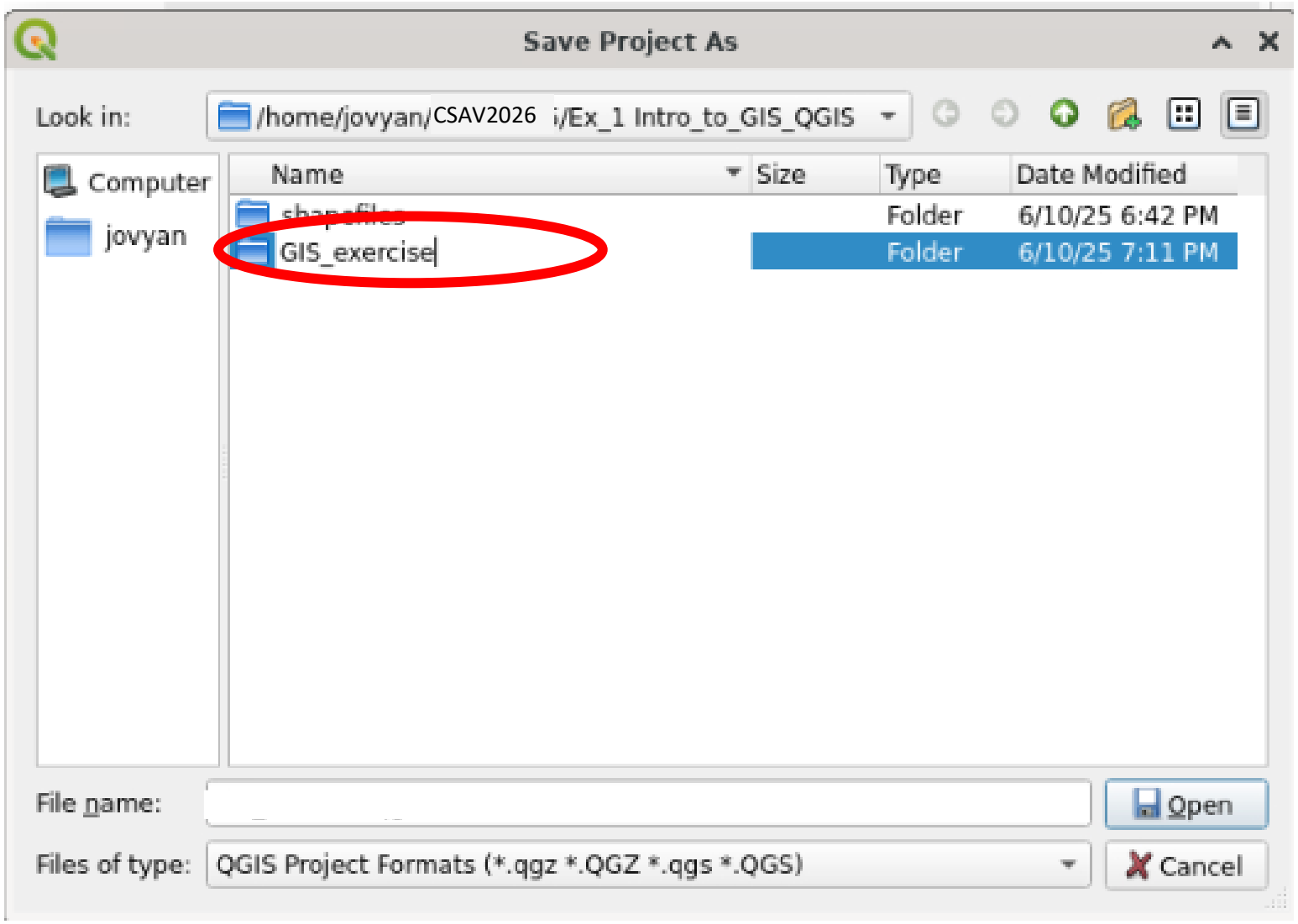

- Navigate to the “CSAV2026” folder



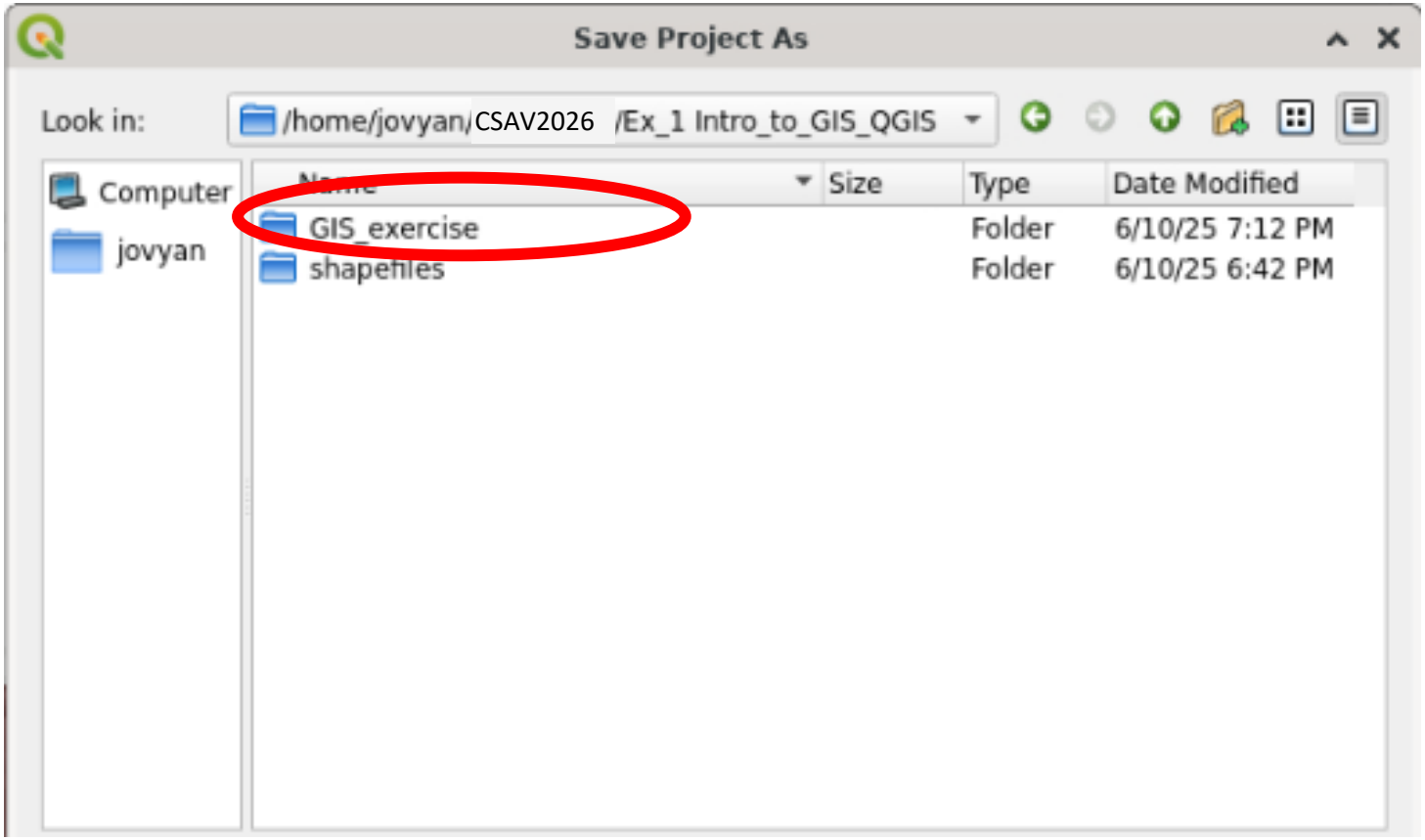

- Navigate to the “CSAV2025/Ex_1_Intro_to_GIS_QGIS” folder



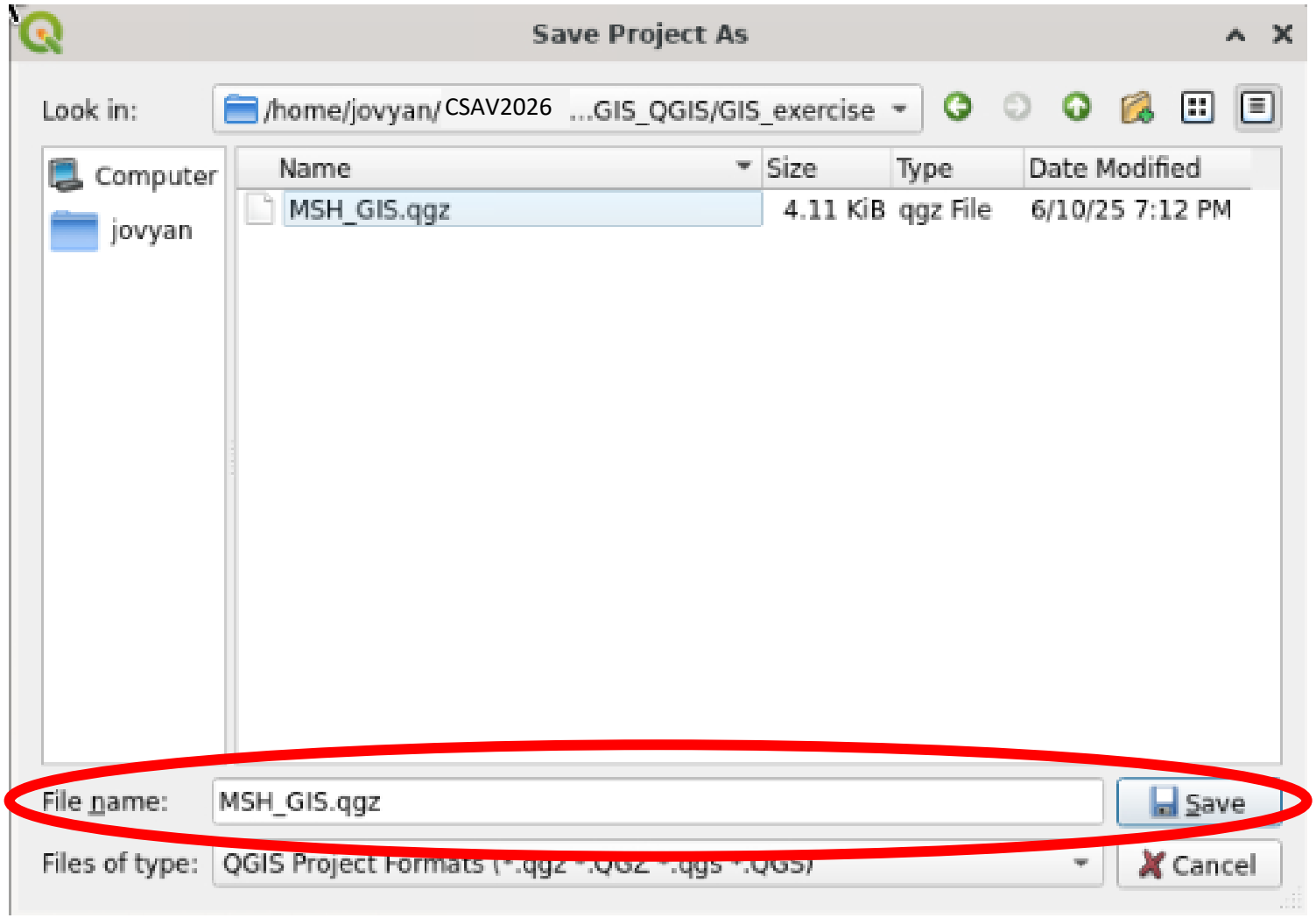

- Click “Create New Folder”



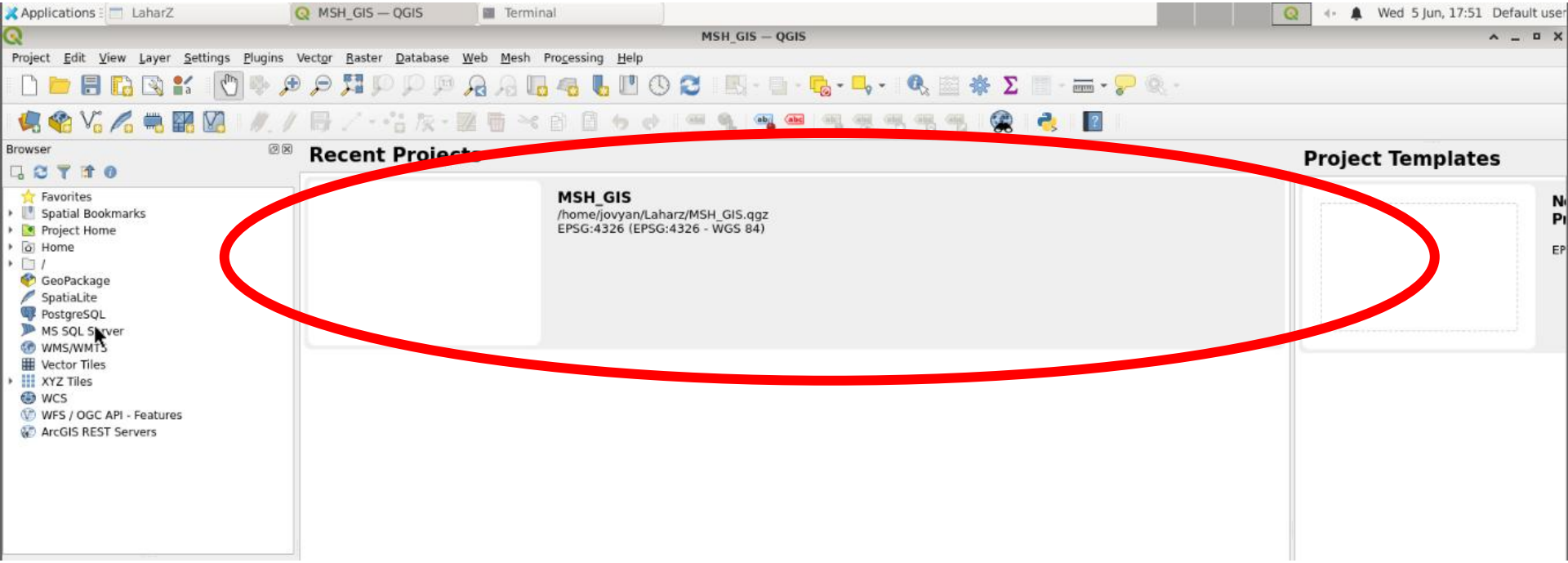

- Name the new folder “GIS_exercise” or something similar




- Double click the new folder “GIS_exercise”



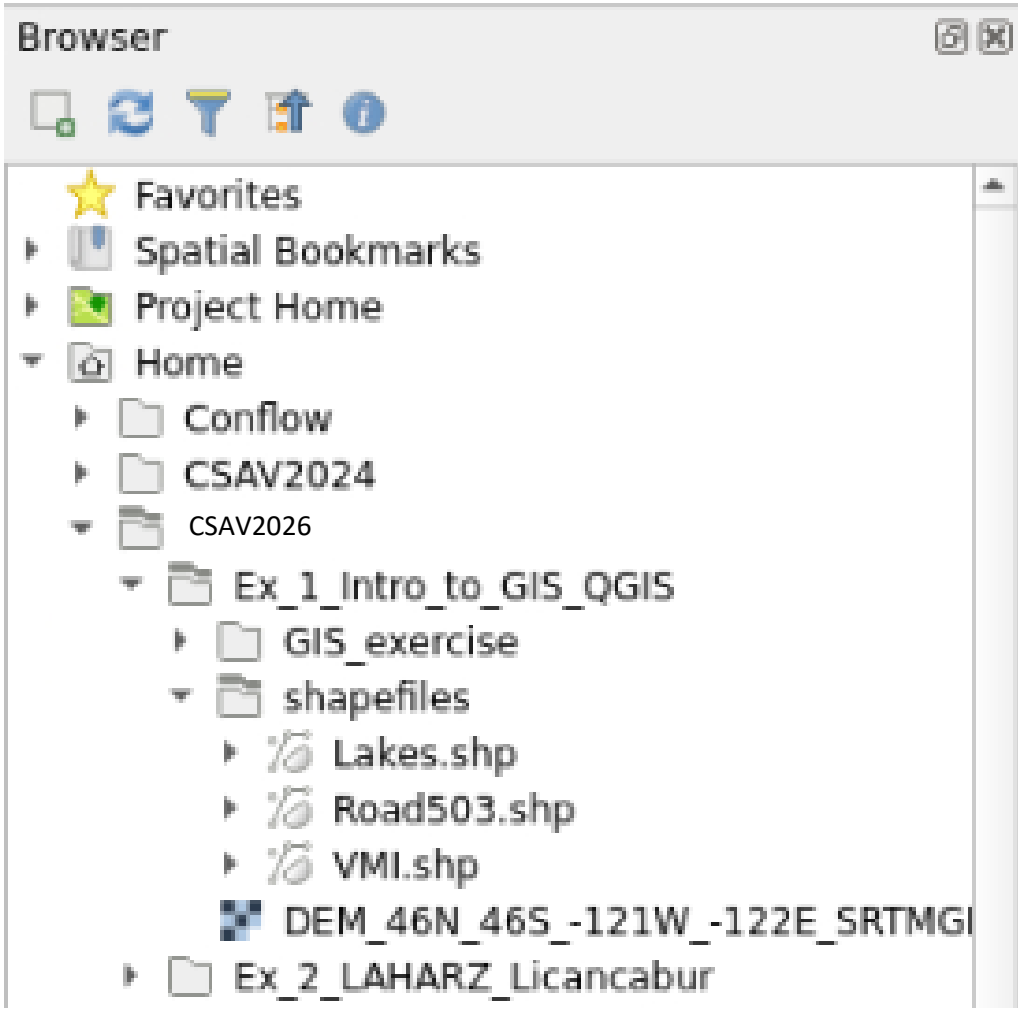

- In the “File name” box, type the name of your project
- I named this “MSH_GIS”
- Click “Save”

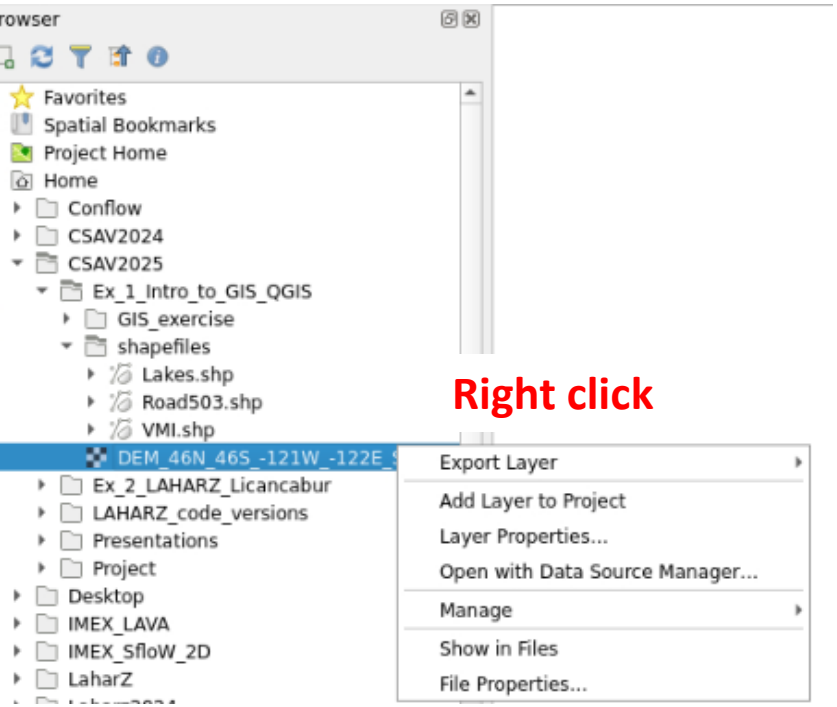


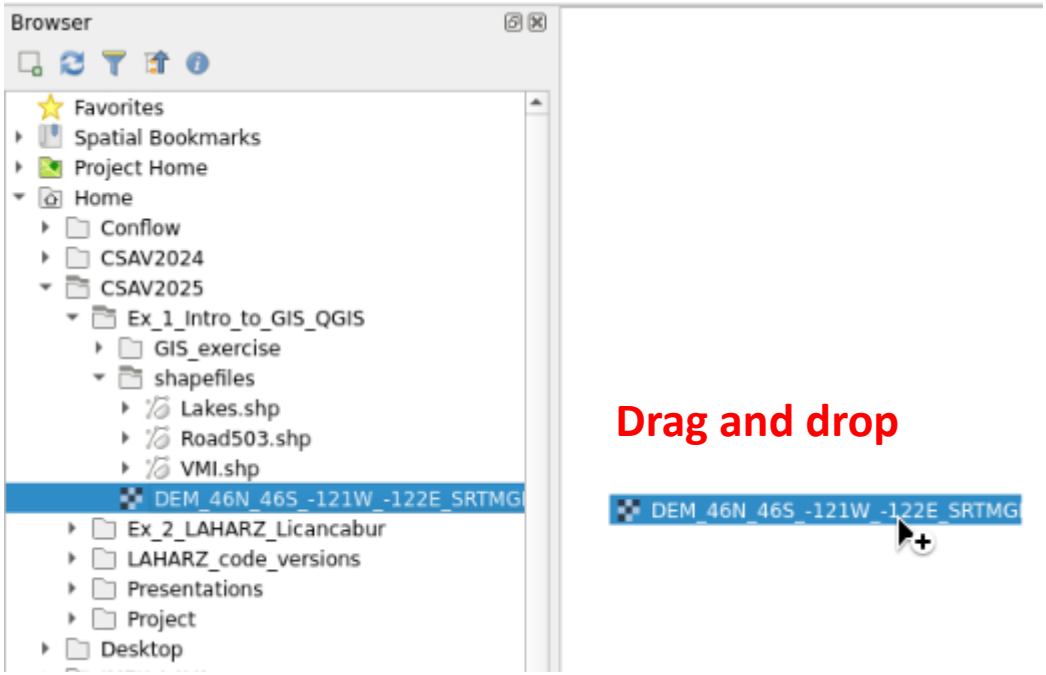

- Click on the new project

# Navigate to files

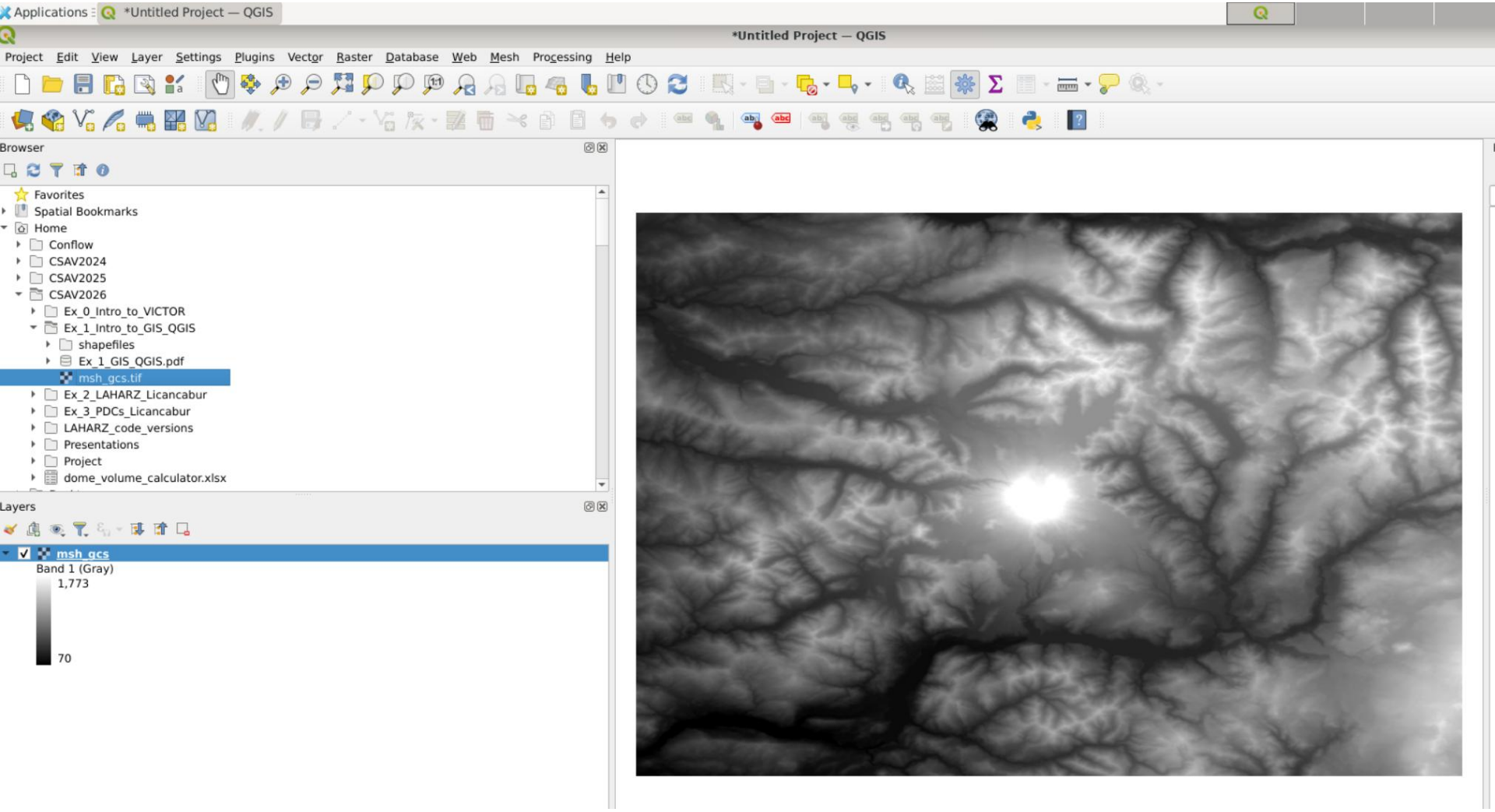
- In the “Browser” panel of QGIS, navigate to the CSAV2026/Ex_1_Intro_to_GIS_QGIS folder
- In this folder are :
  - The “GIS_exercise” folder you just made
  - The DEM you downloaded from VICTOR
  - A folder with lake, road, and the volcano monitoring instrument (VMI) shapefiles



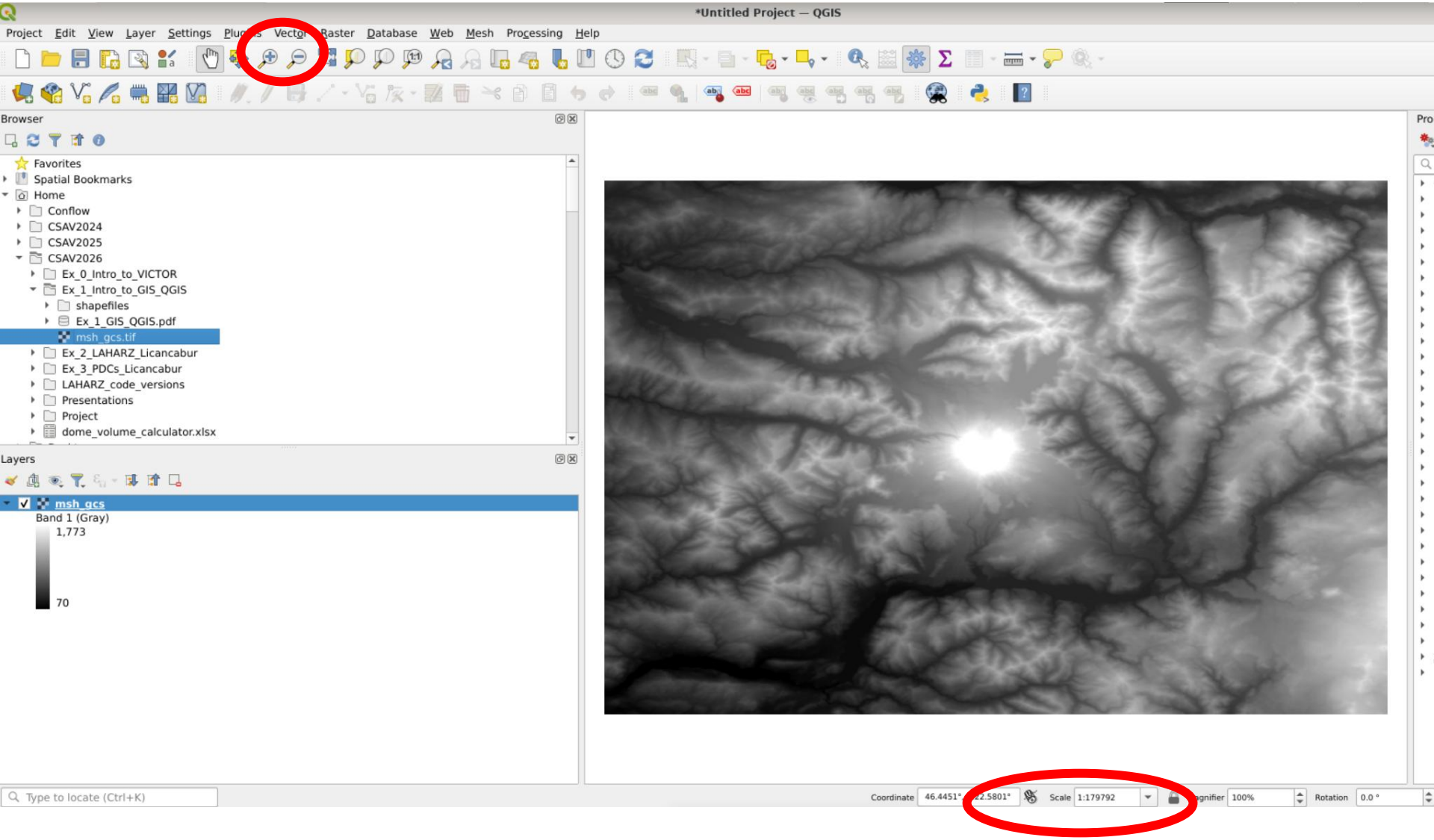


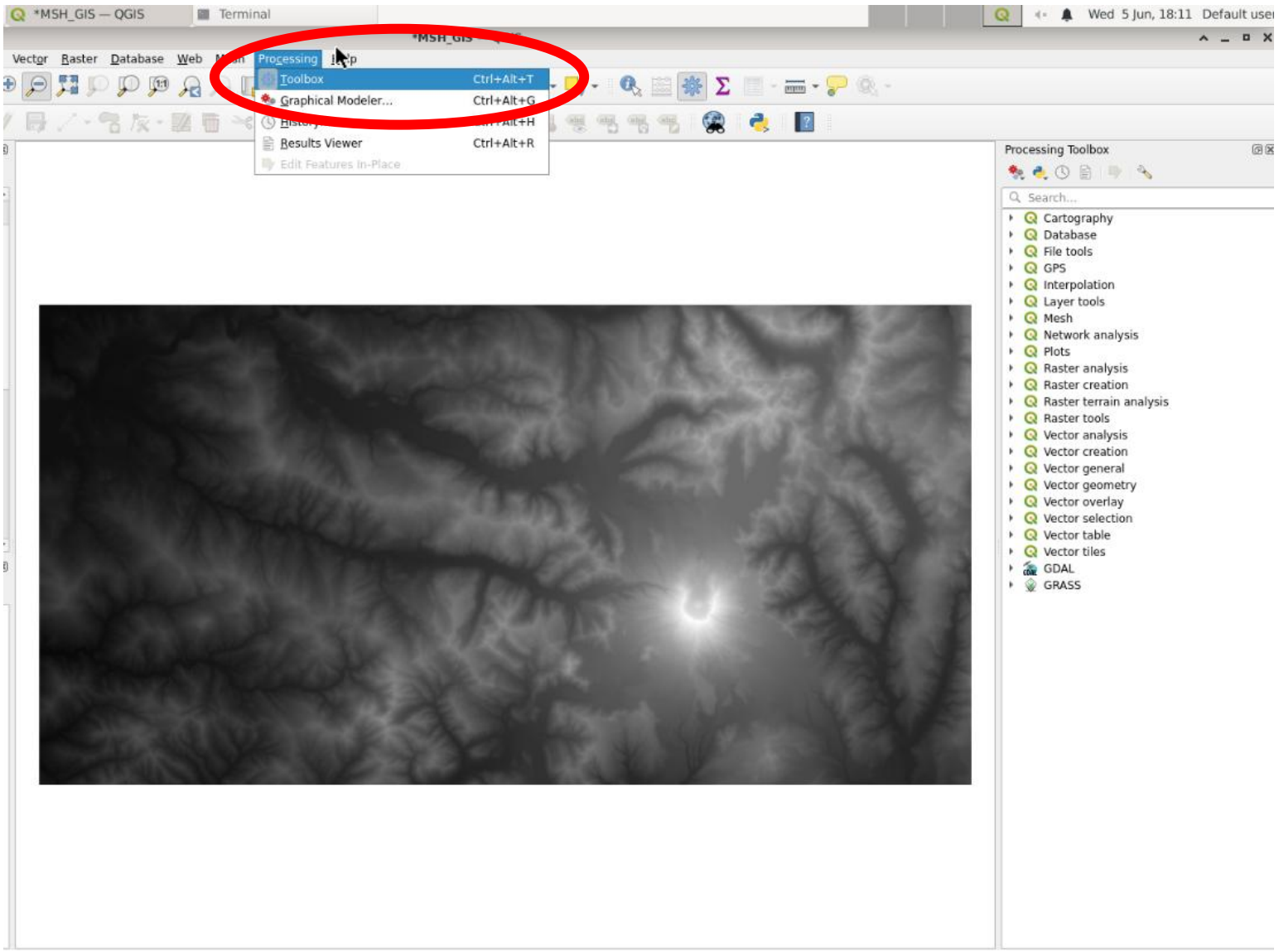

- Two ways to add layers:
  - Right click on the DEM file and choose “Add Layer to Project”
  - Drag the DEM layer and drop onto the workspace




- You should see the MSH DEM on the screen

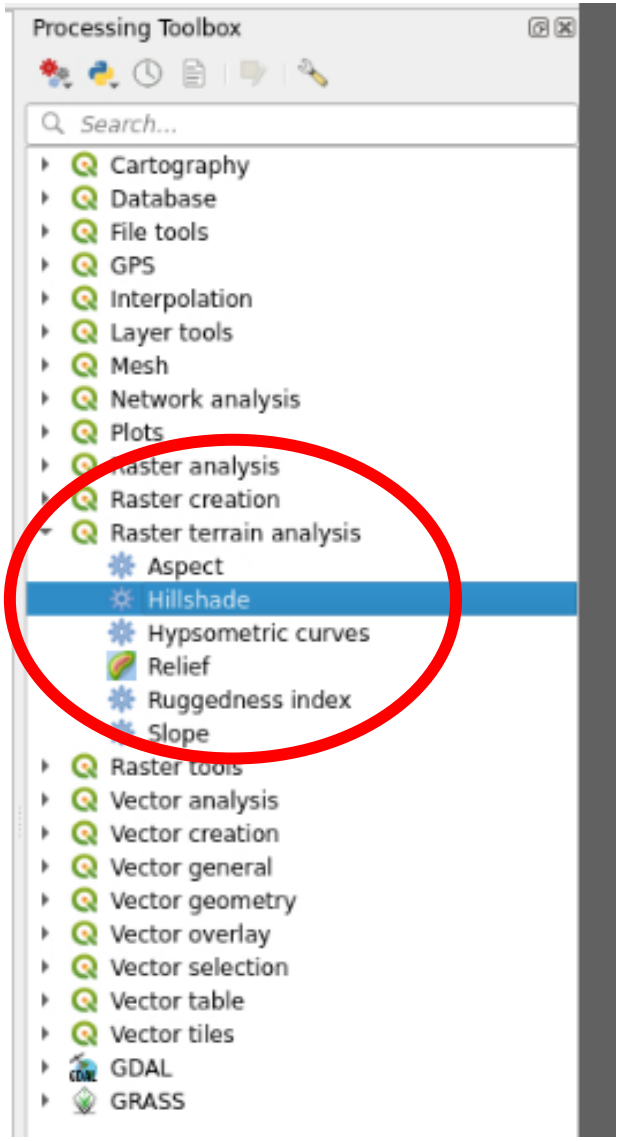



- You can zoom in and out 3 ways:
  - Use the mouse wheel
  - Click the magnifying glass icon, then click on the map
  - Edit the map scale manually at the bottom of the screen. Try 1:90000




- Open the Processing Toolbox

- Let’s make a hillshade
  - A hillshade (or shaded-relief model) is a rendition of shadows across a DEM that helps our brains interpret what we see in DEMs.
  - Click the “Raster terrain analysis” menu
  - Double click the “Hillshade” tool
  - You can also use the “Search” field to search for tools!

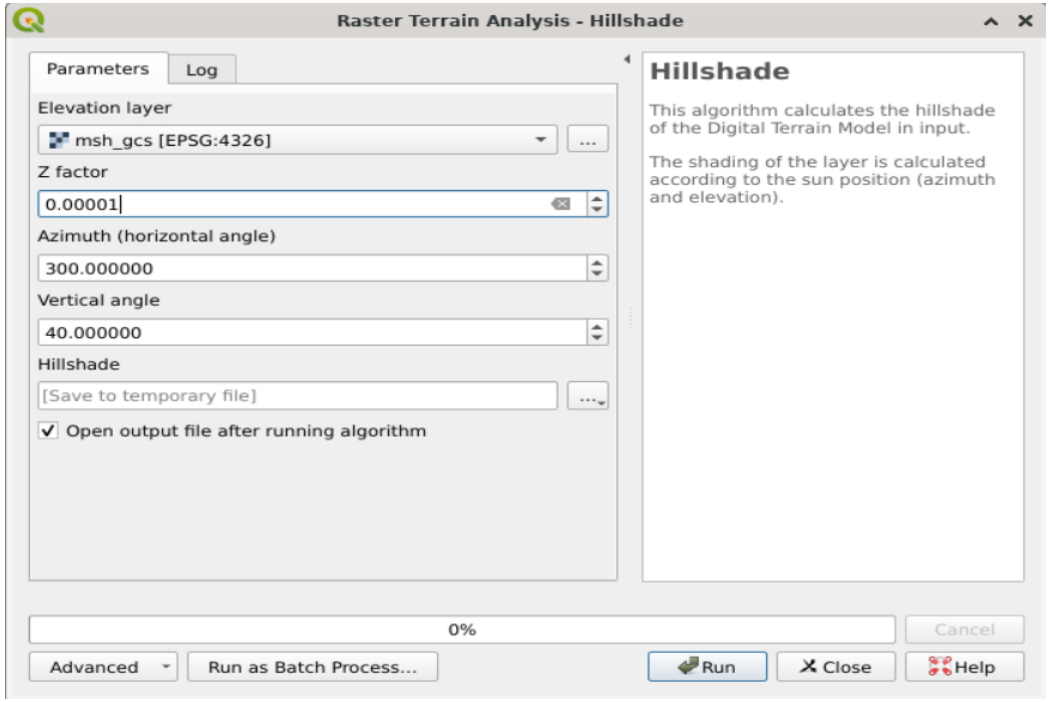


# Let's make a hillshade

This tool lets you model a "sun" and the "shadows" it would produce on your DEM

Use these parameters

- Z factor is vertical exaggeration (if your z units differ from your x and y units)
- We are using geographic coordinates (degrees), so our z units (meters) ARE different from x, y (decimal degrees)
- Enter 0.00001 as Z factor
- The azimuth is where the sun is relative to your map. The human brain expects the shadows to be modeled with the sun coming from the upper left (300°)
- The vertical angle is how high in the sky the modeled sun is. We use the default 40°



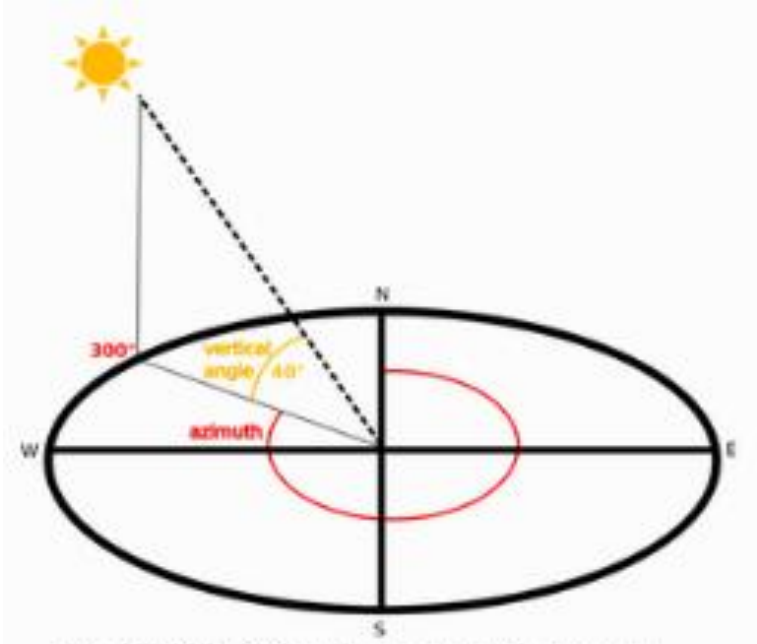
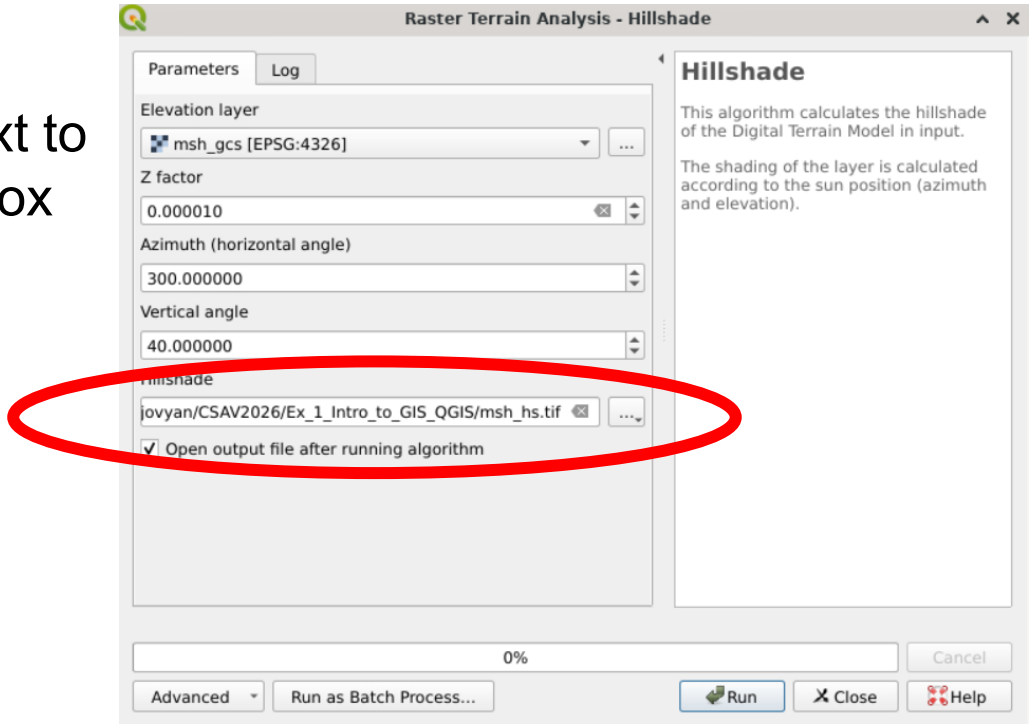
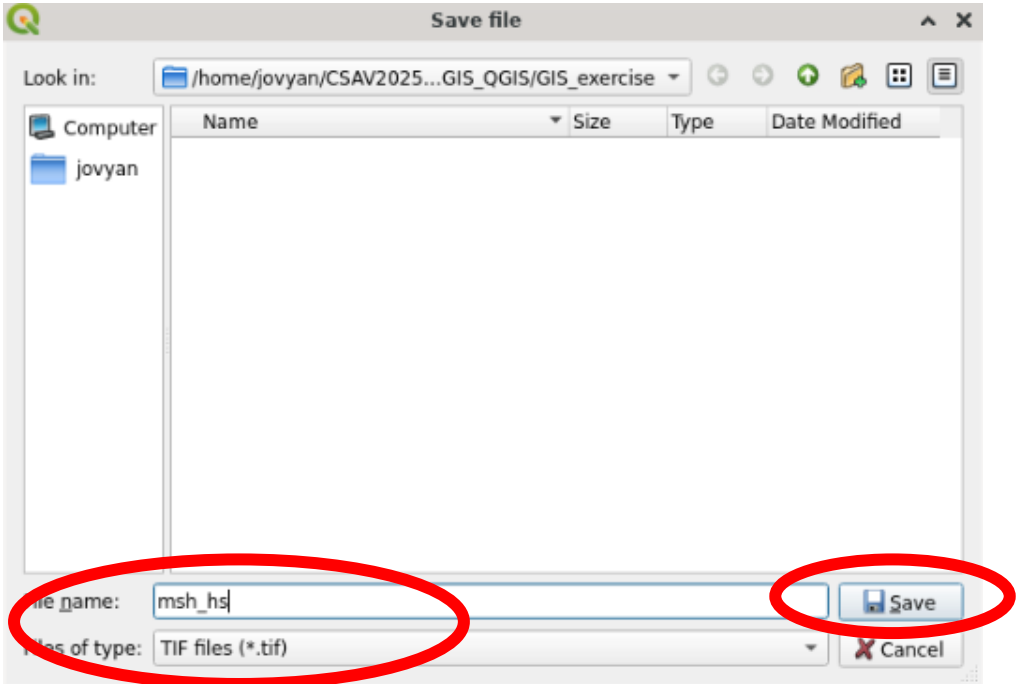



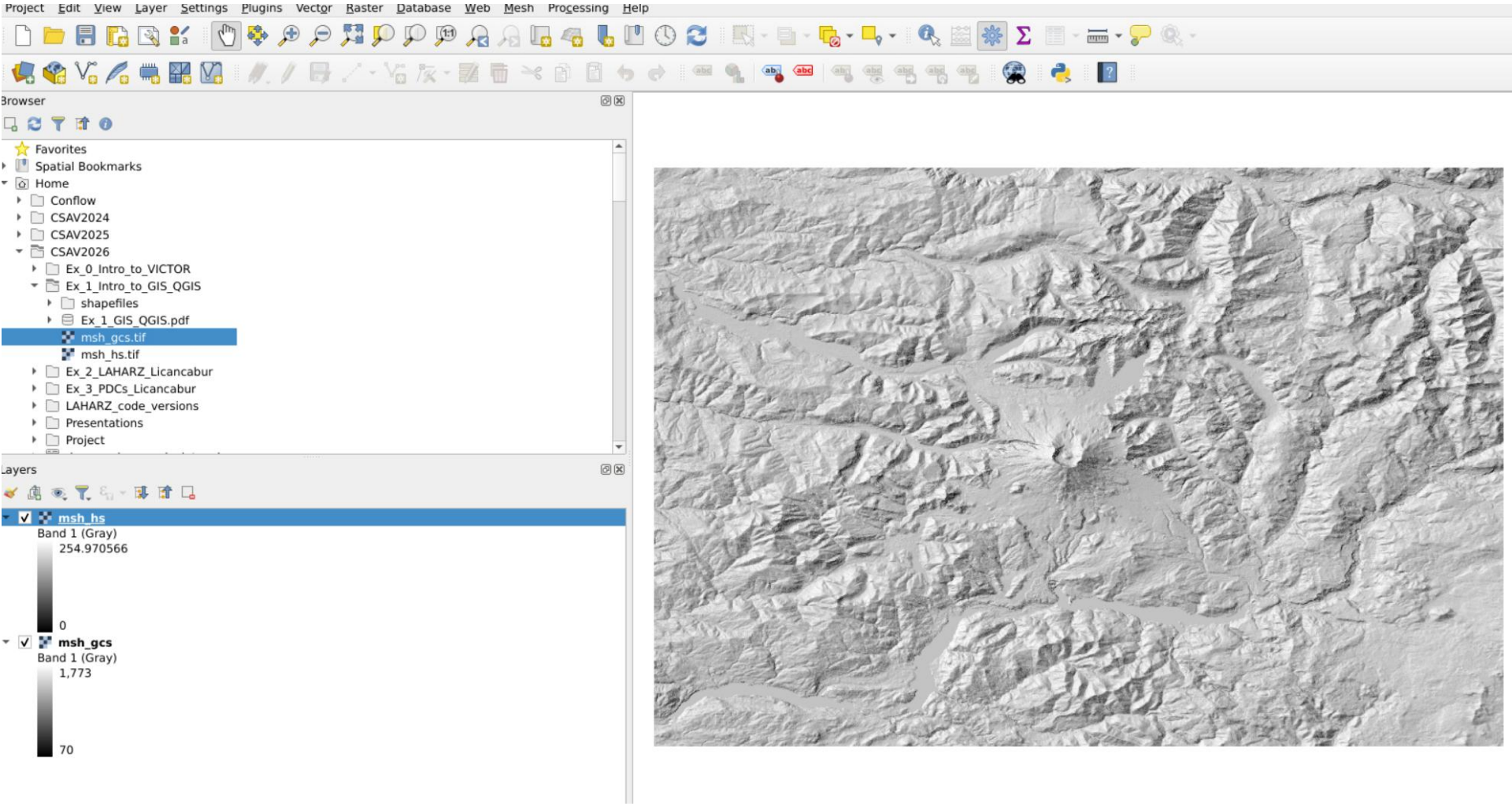

Fig. 28.32 *Azimuth and vertical angle*

Let's make a hillshade

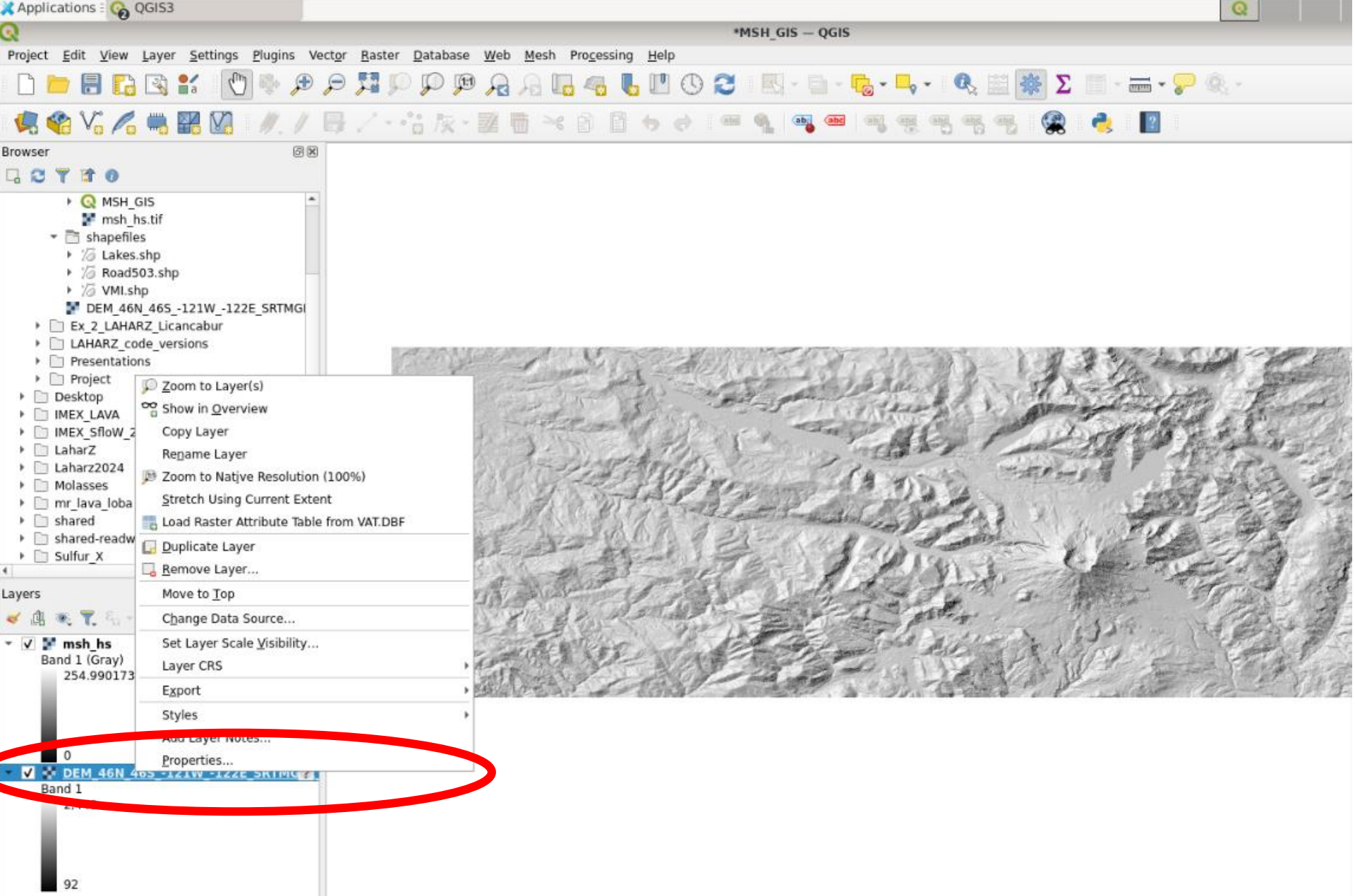
- Click the 3 dot menu next to the Hillshade filename box and select "Save to File"
- Navigate to the GIS_exercise folder
- Enter a filename. I used "msh_hs"
- Click "Save"
- Click "Run"



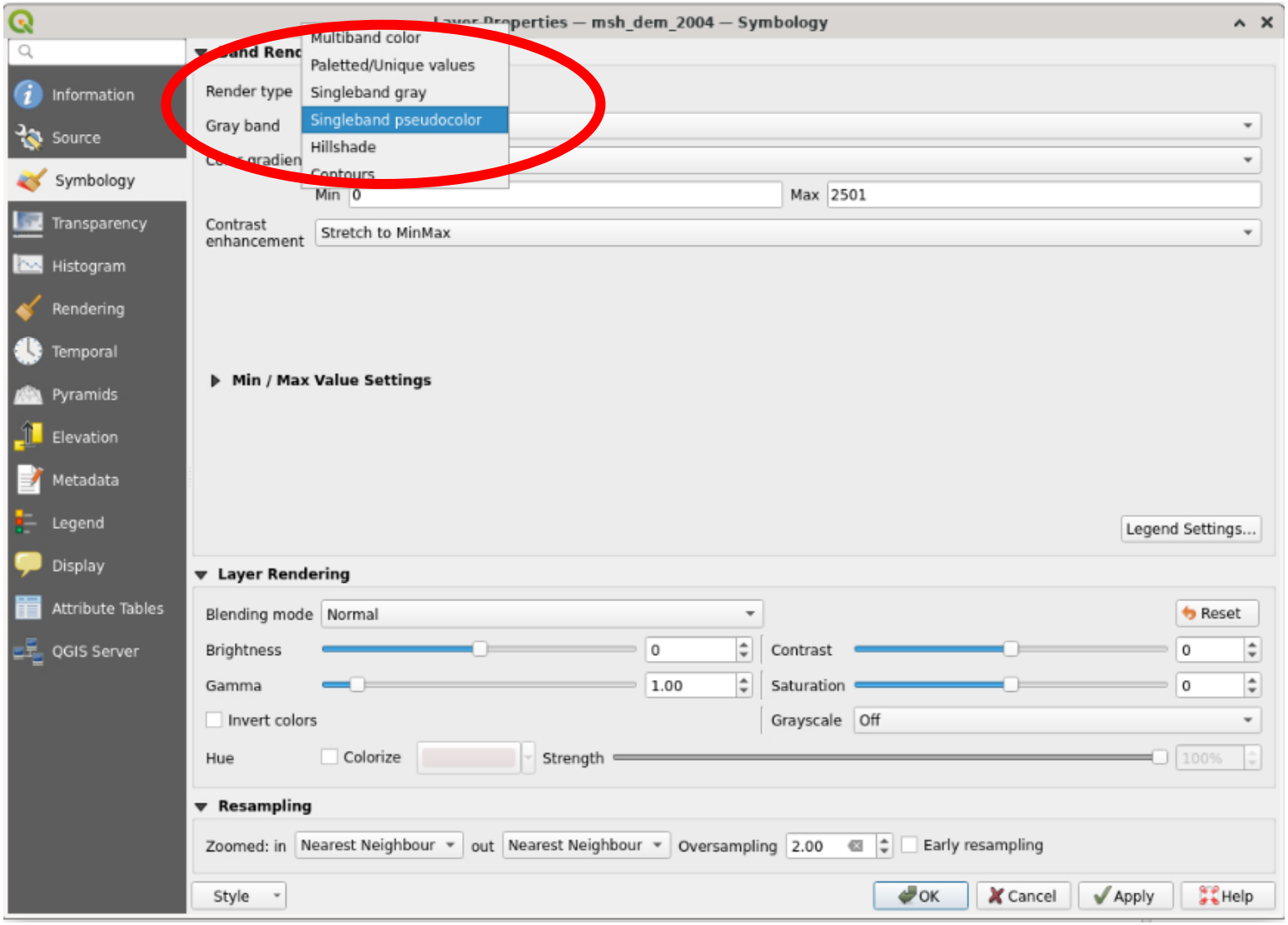





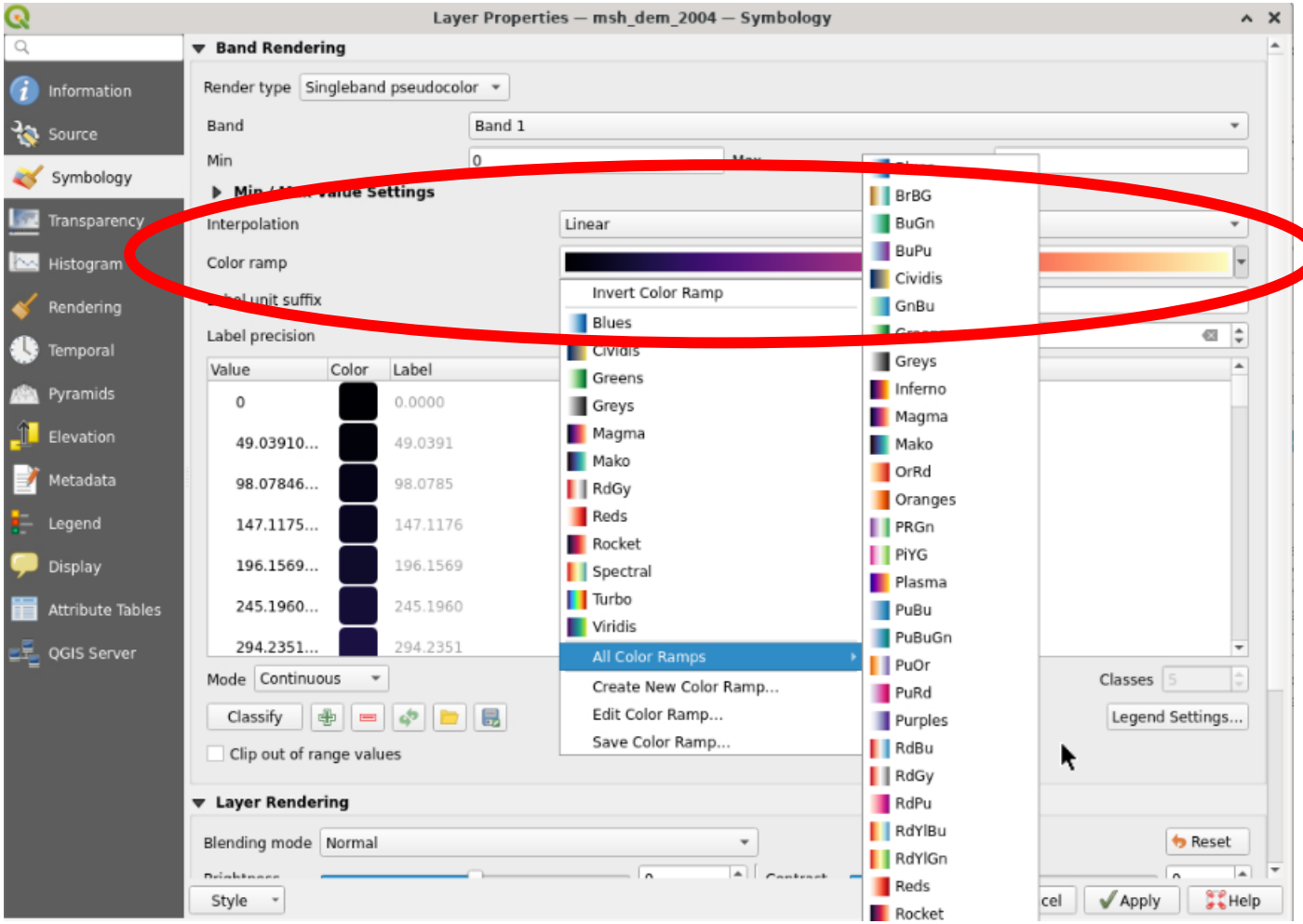

- We can change the symbology of the DEM and the transparency of the hillshade if we want
- Hillshades (shaded relief) use shades of grey 0-254. DEMs use a stretched color spectrum from lowest to highest elevation



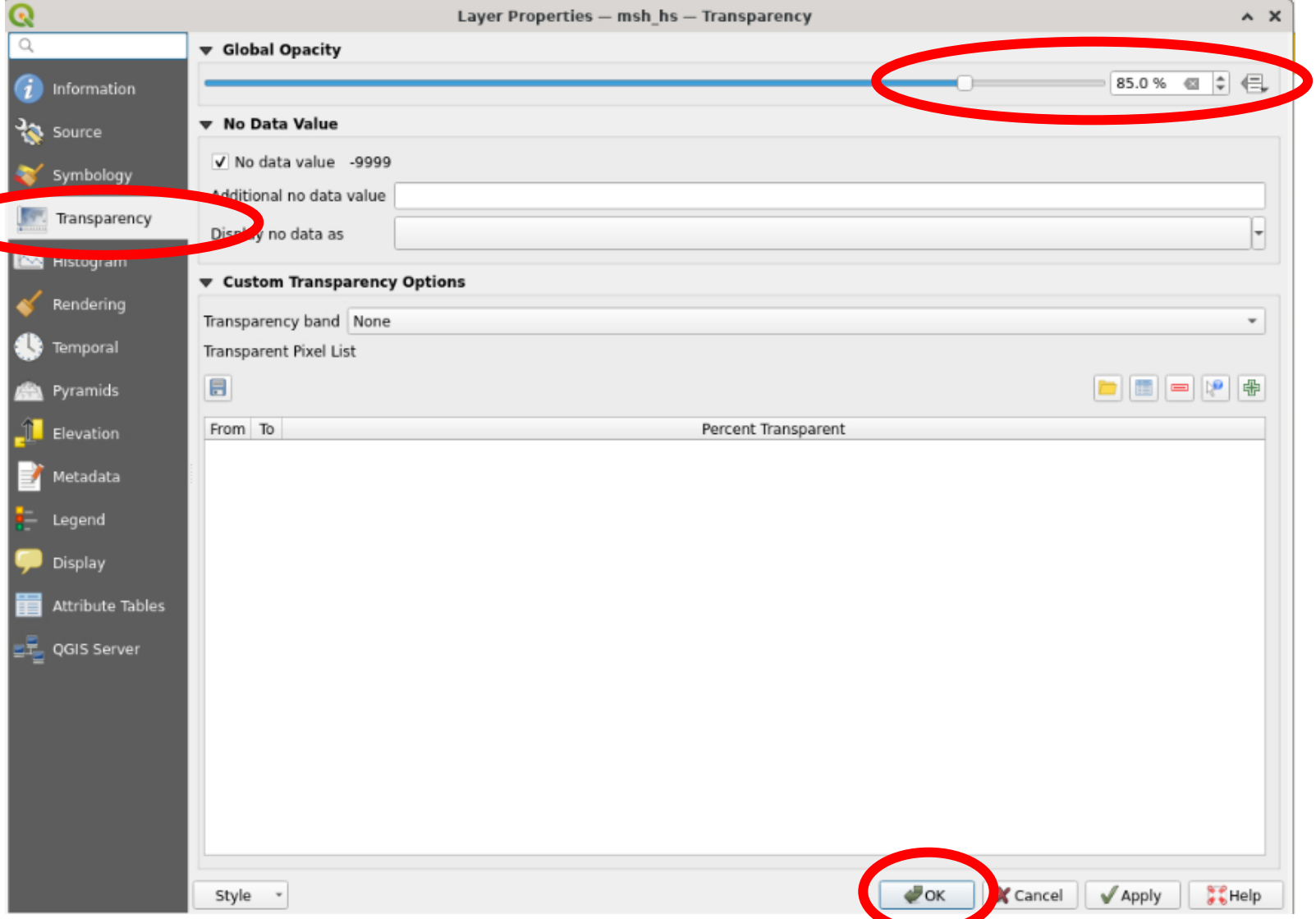

- Double click on the DEM layer (msh_gcs) to open the layer properties
- OR right click on the DEM layer and choose “Properties”



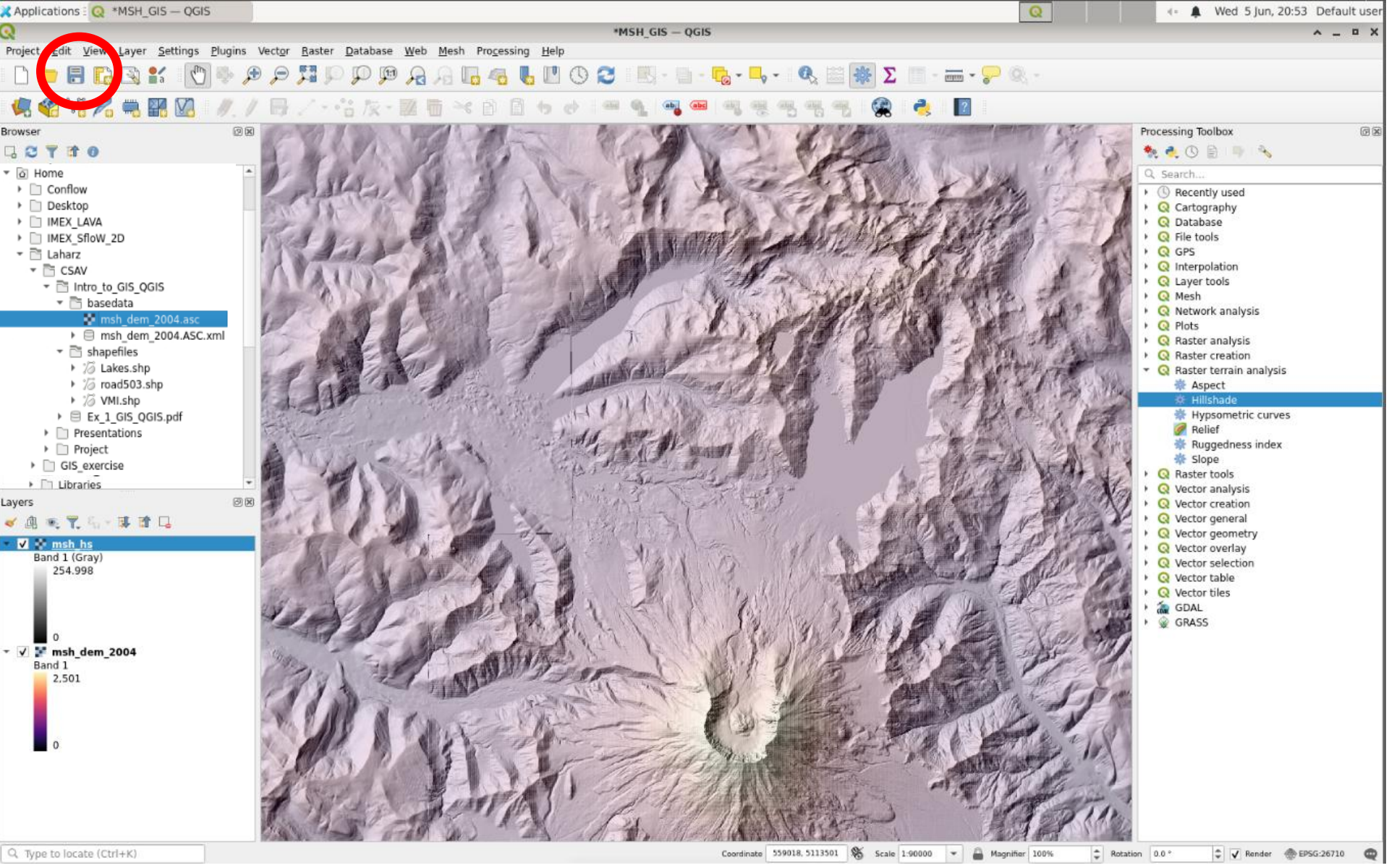

- Change the “Render type” to “Singleband pseudocolor”



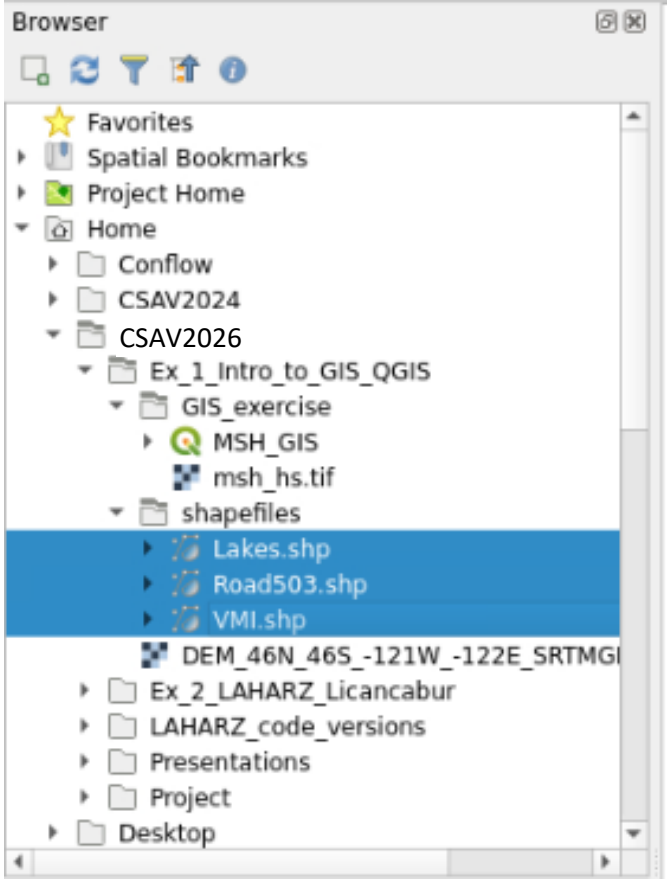

- Choose a color ramp
- I chose the “Magma” color ramp
- Click “Ok”



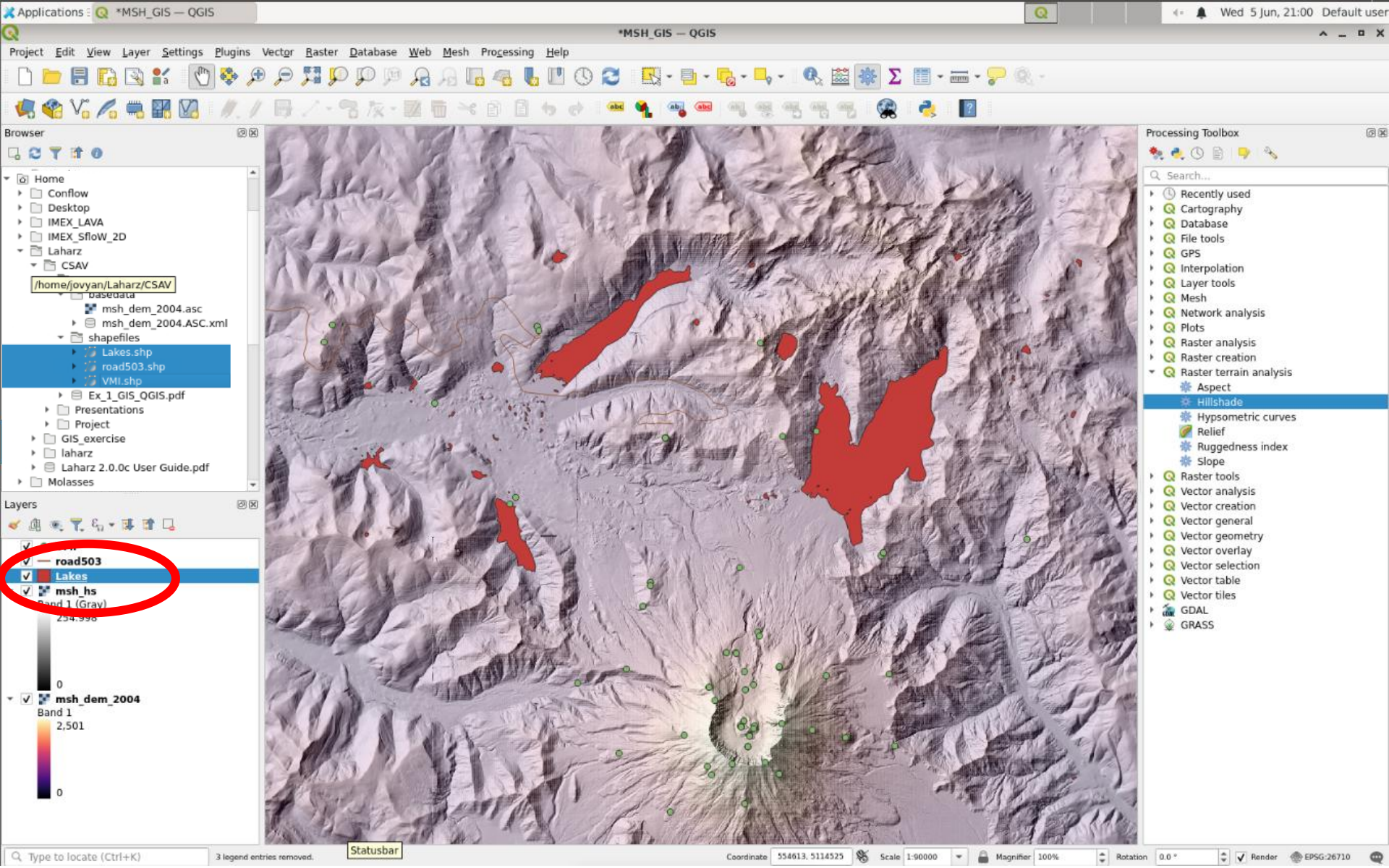

- Now let's make the hillshade slightly transparent
- Right click on "msh_hs" and select "Properties"
- Then select "Transparency" from the left menu
- Change the "Global opacity" to something less than 100% (I chose 85%)
- Click "OK"



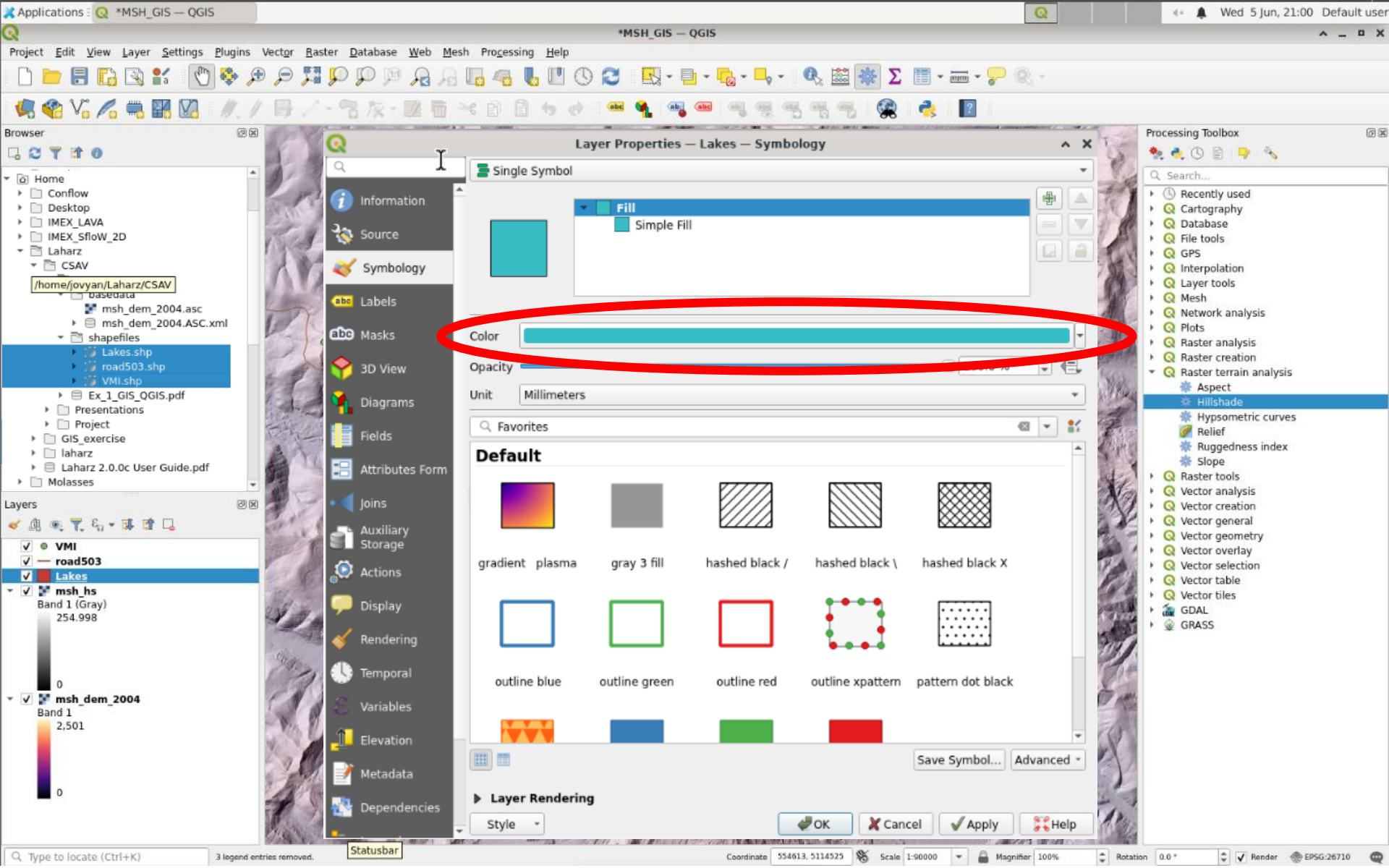

- Now's a good time to save your project!
- Click the "Save" icon in the upper left



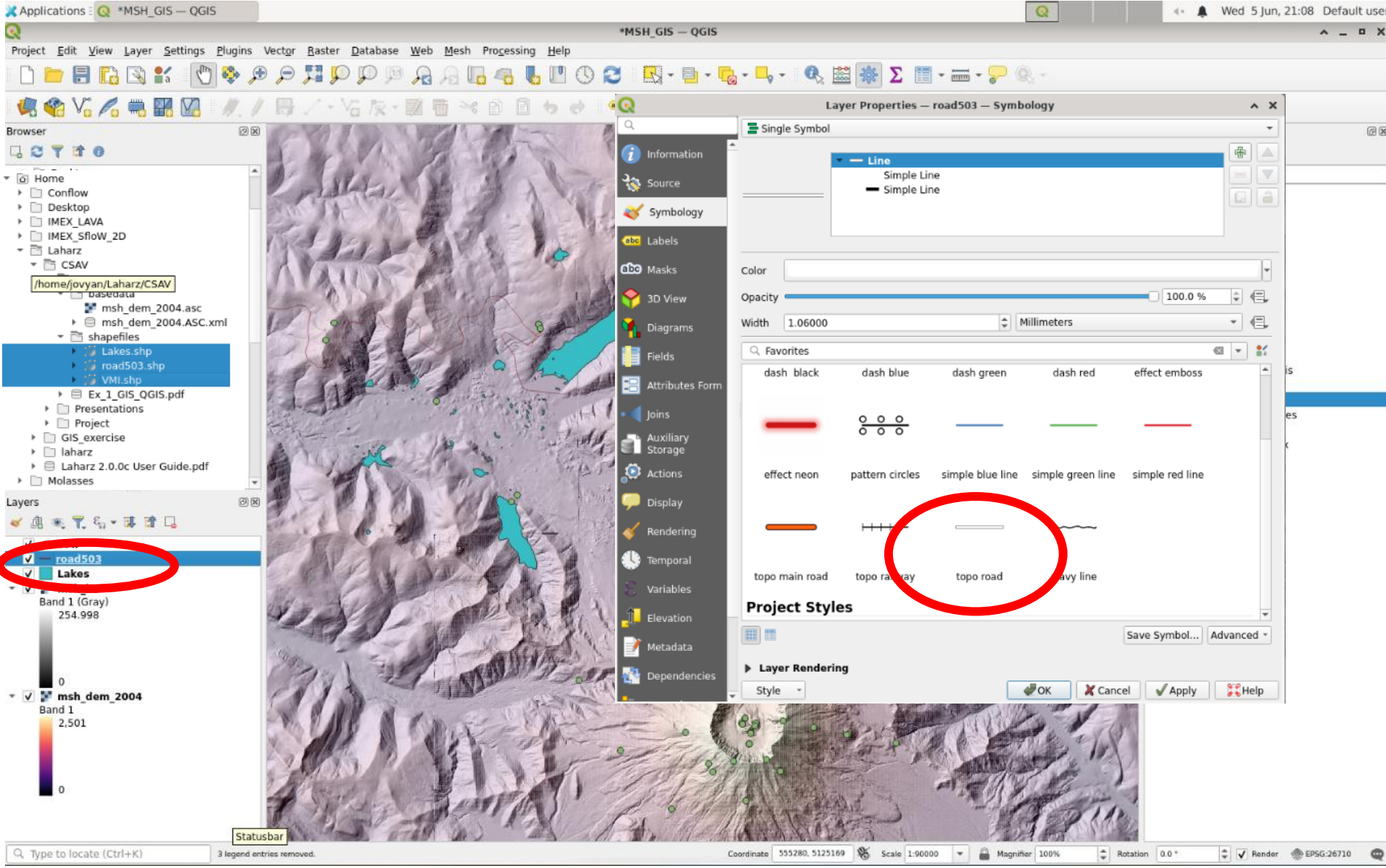

- Add data: Lakes, road503, and VMI shapefiles




- The default symbology for these layers isn't very nice…
- Let's fix it!
- Double click on the "Lakes" layer to open the Layer Properties



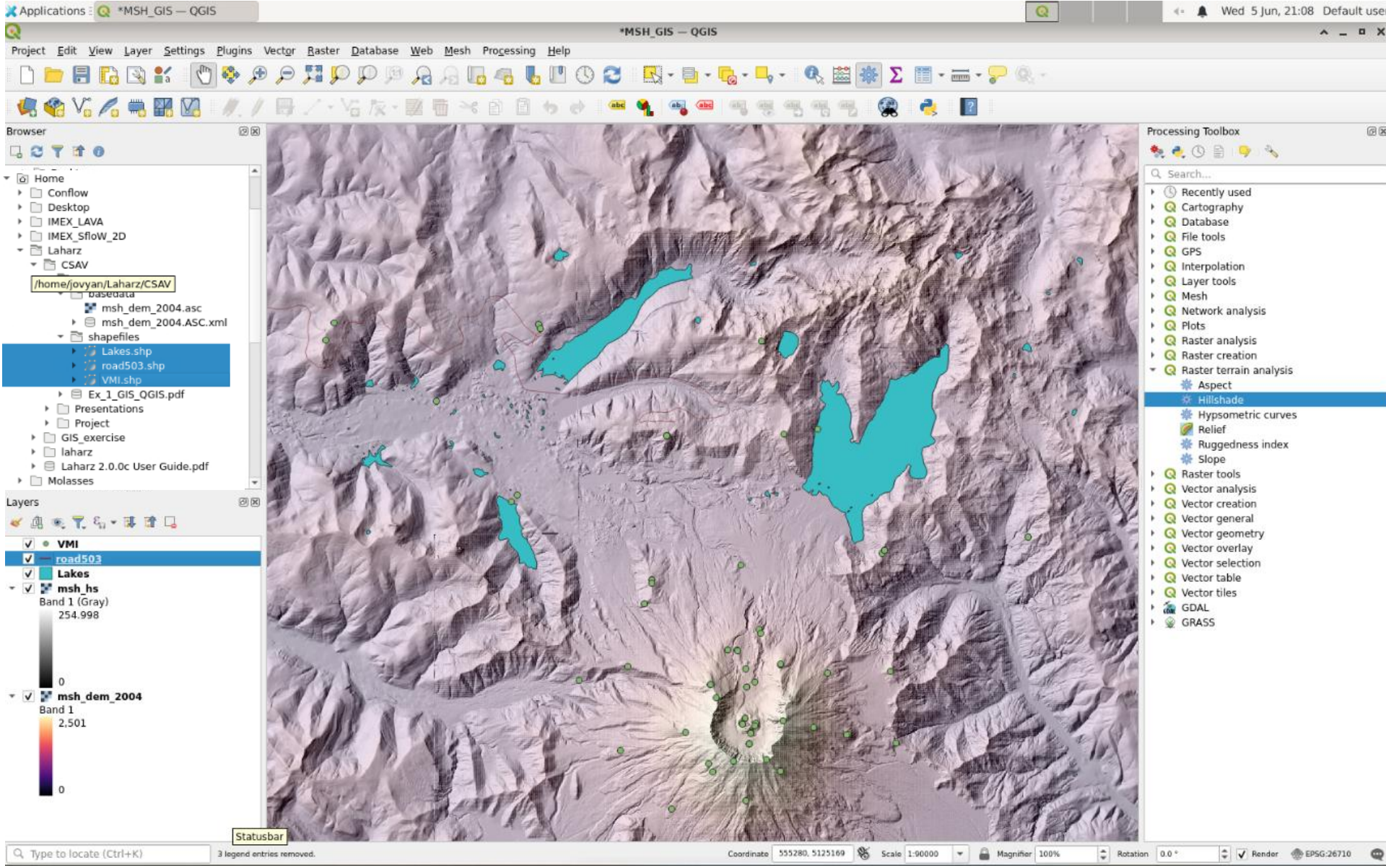

- Change the lake color to a lake color
- Use the drop-down menu labelled “color” and choose a color
- Click “Ok”



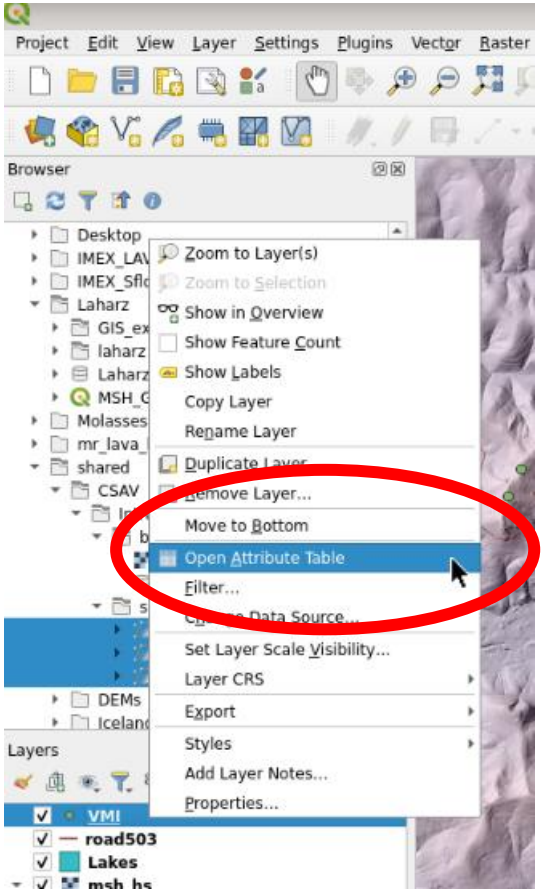

- The road is pretty hard to see. Let's fix it.
- Open the layer properties for "road503" and change the symbology. I chose "topo road"
- Click "Ok"

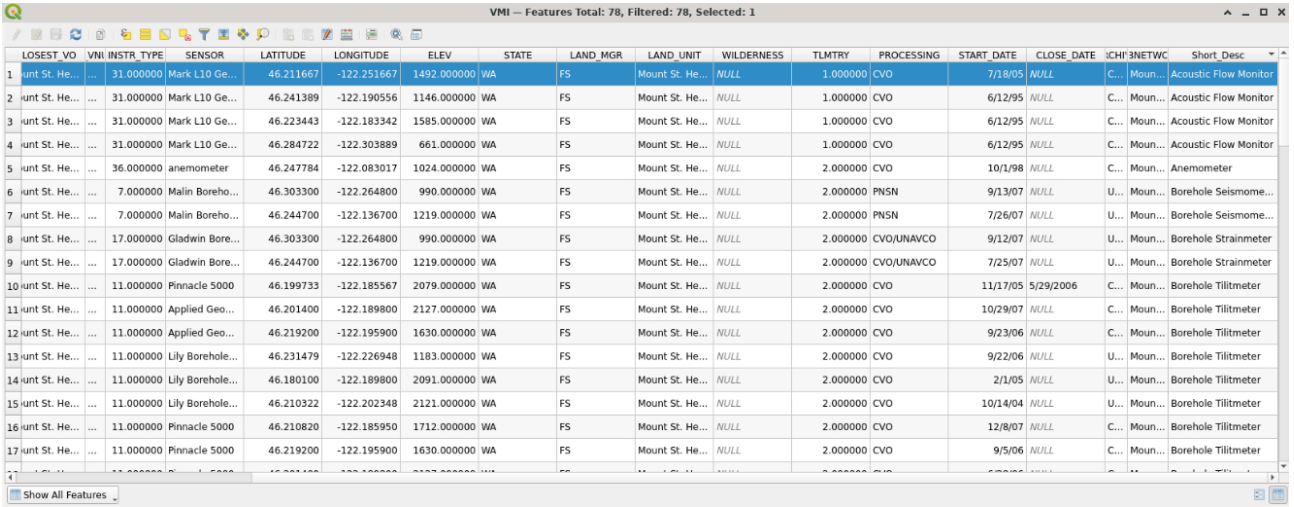
# Exercise Goals

- **View Instrument attribute table**
- Select instruments by drawing rectangle
- Select instruments by their attributes
- Assign colors to instrument type
- Make a map and add a legend to describe the meaning of colors

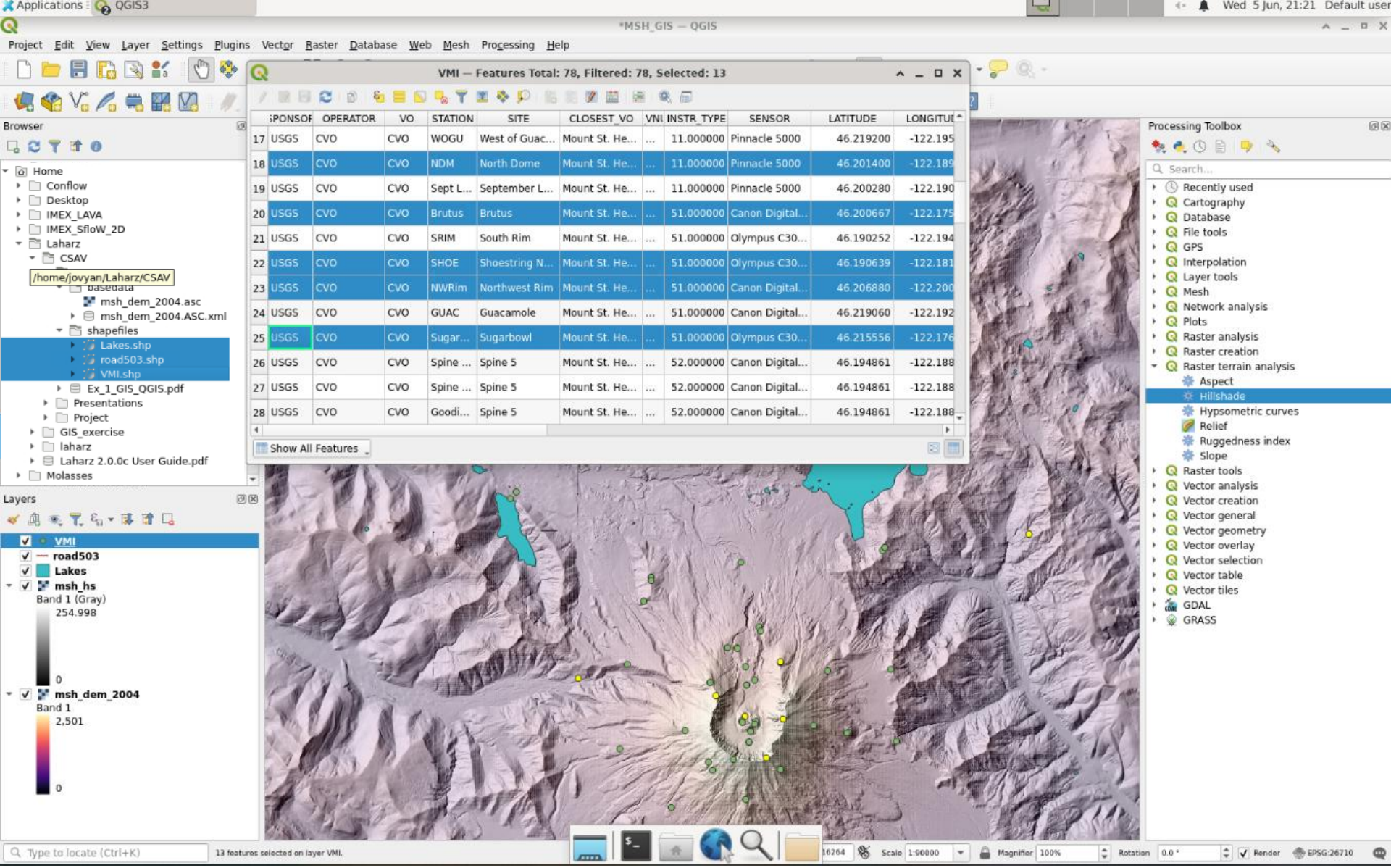



- The last layer is “VMI” or “volcano monitoring instruments”. It shows the network of instruments of different types.
- The default symbols are all the same, so we can’t tell what type of instruments these are. Let’s fix that!



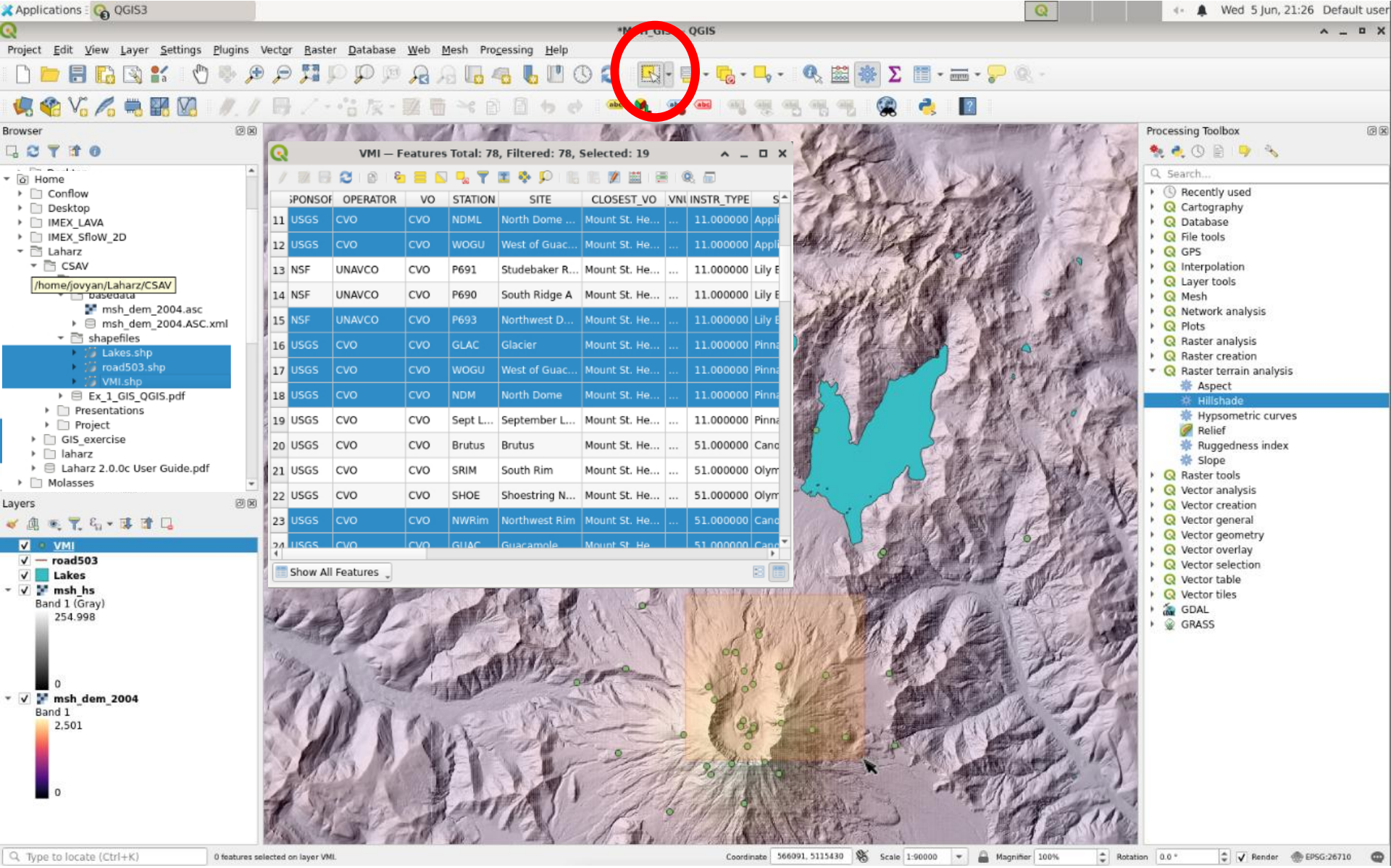

- Right click on “VMI” and choose “Open Attribute Table”
- We can see all kinds of information about the sensors, including the sensor type (“Short_Desc” column all the way to the right)



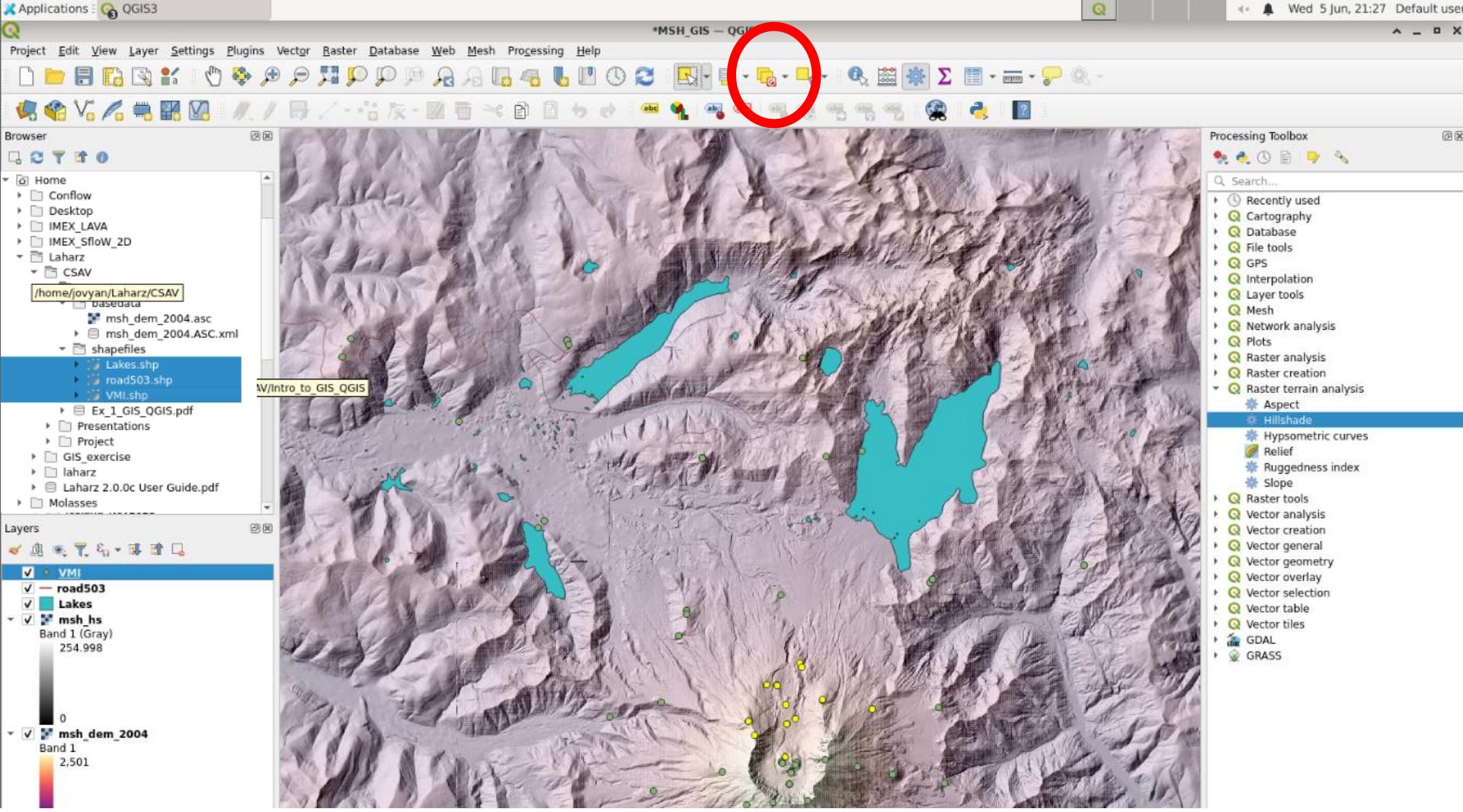

- When you select rows in the attribute table (Ctrl+Click the row number), you will see those instruments turn yellow on the map
- Close the attribute table

# Exercise Goals

- View Instrument attribute table
- **Select instruments by a drawing rectangle**
- Select instruments by their attributes
- Assign colors to instrument type
- Make a map and add a legend to describe the meaning of colors



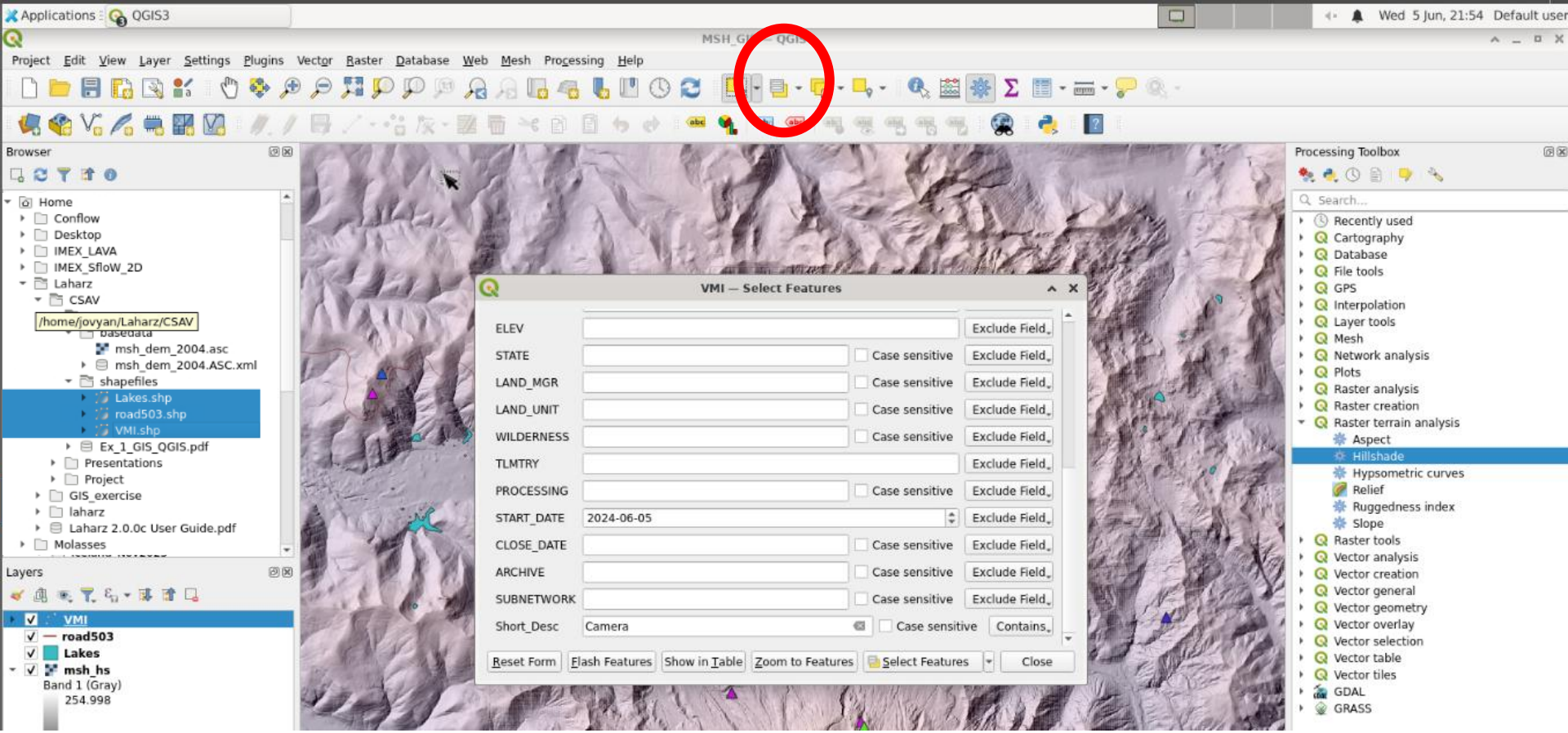

- Choose the “Select” tool from the menu
- Draw a rectangle around some of the VMI stations
- The selected stations will highlight in yellow
- If you open the attribute table you will see the selected stations highlighted



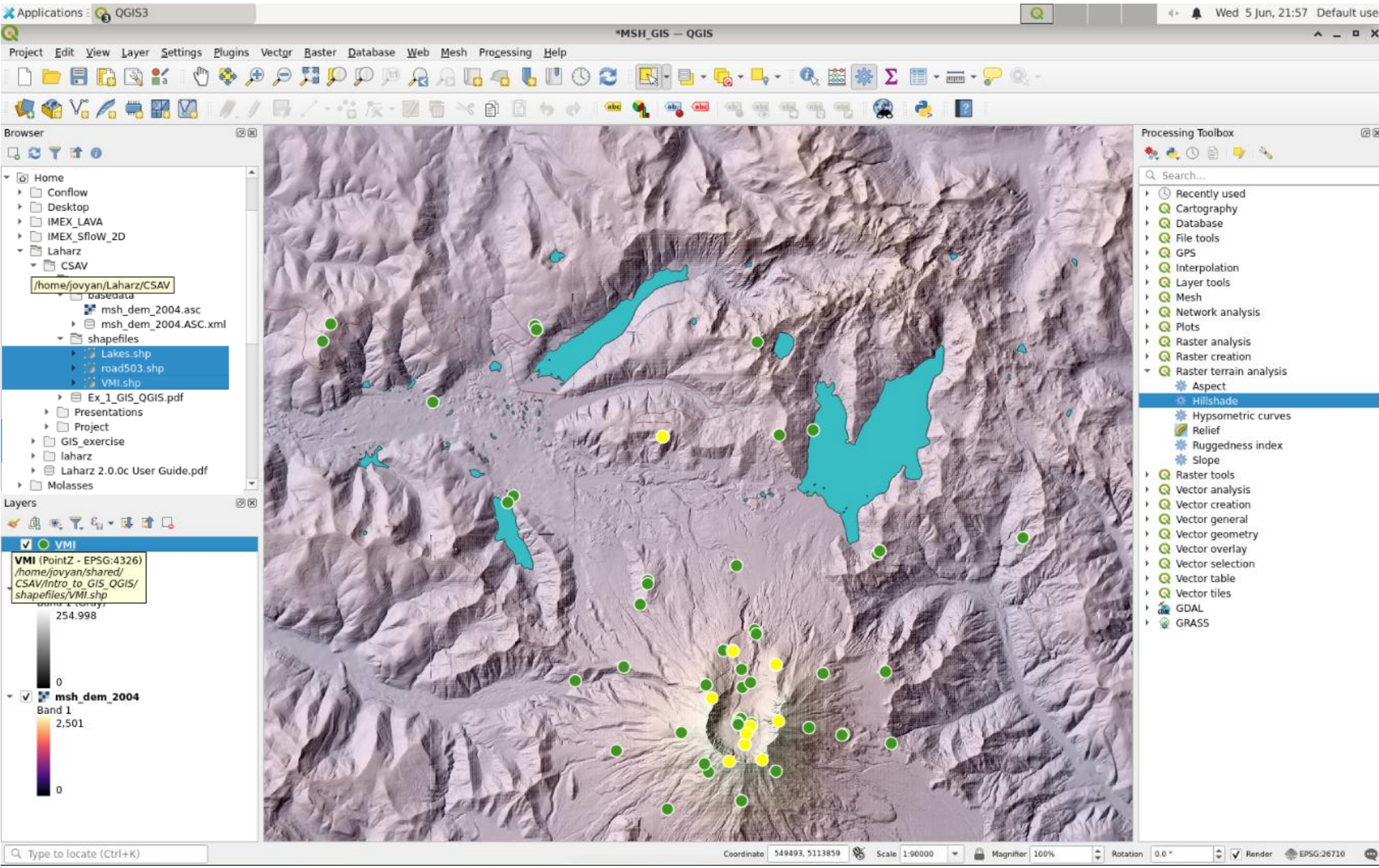

- To clear the selection, click the “Deselect features from all layers” button

# Exercise Goals

- View Instrument attribute table
- Select instruments by a drawing rectangle
- **Select instruments by their attributes**
- Assign colors to instrument type
- Make a map and add a legend to describe the meaning of colors



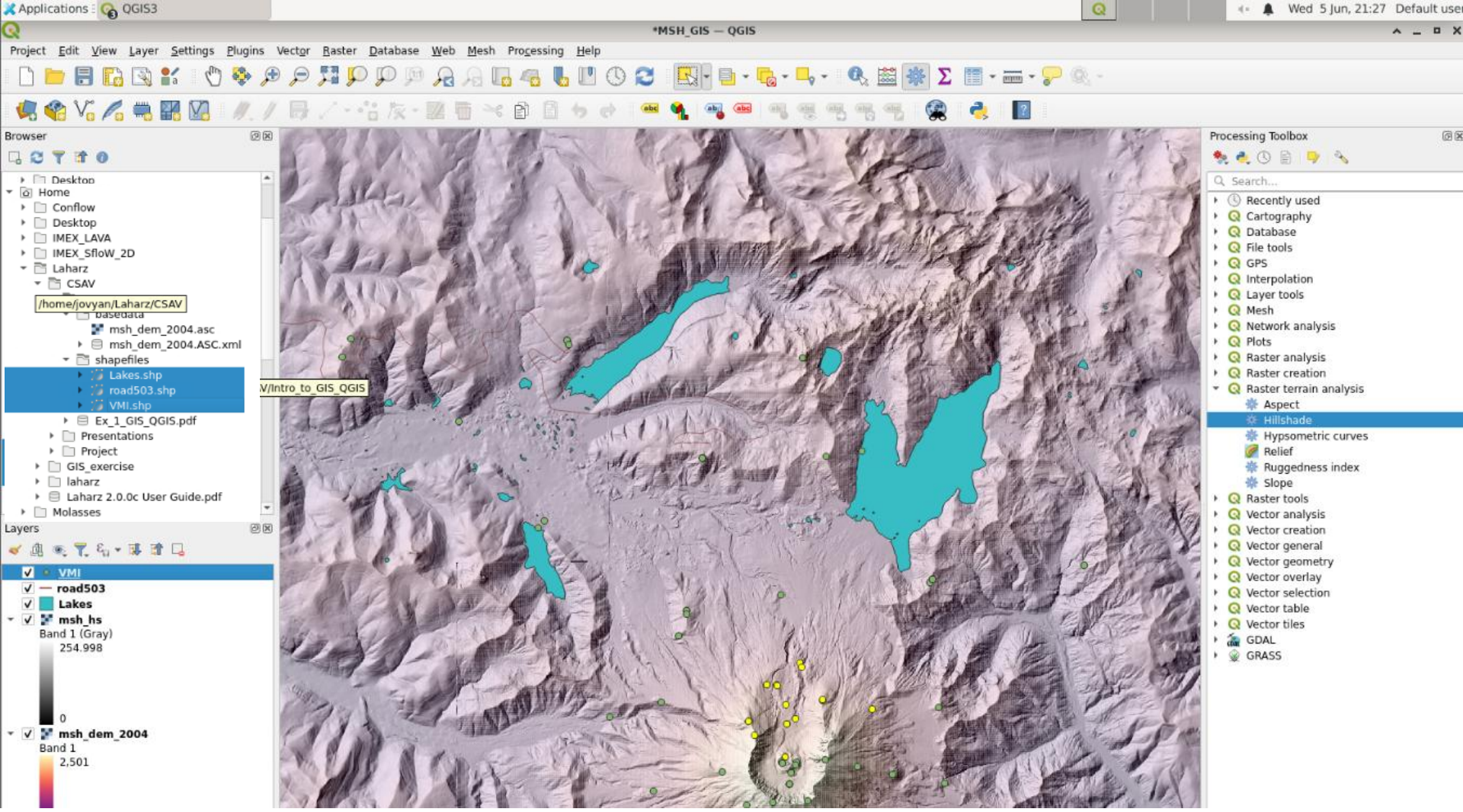

- We can query the attribute table to select certain features based on data from the table
- Select the “Select features by value” tool
- Scroll down until you see “Short_Desc”
- Change the button that says “Exclude Field” to “Contains”
- Type “Camera” in the text box
- Click “Select Features”



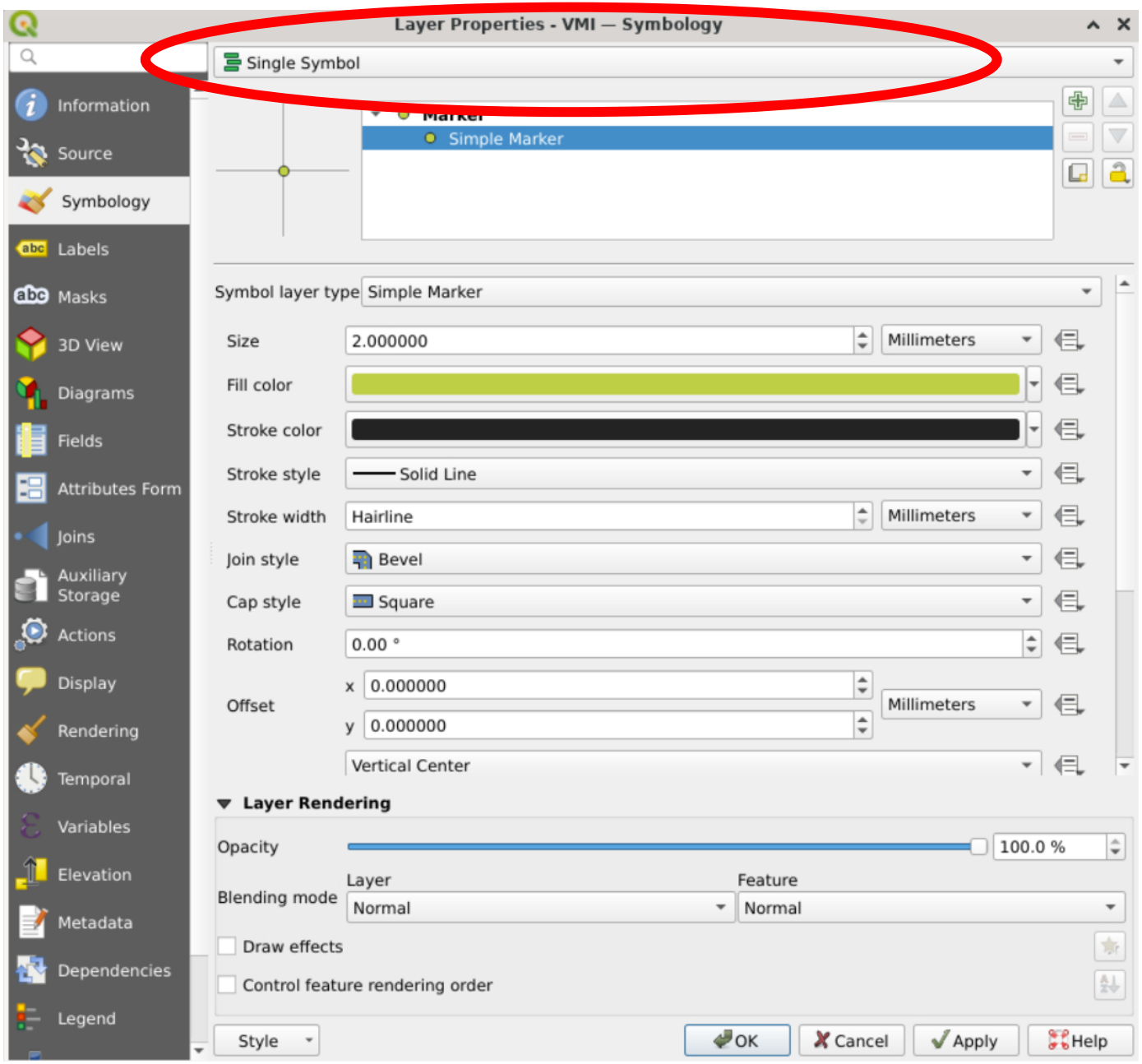

- The camera stations will be highlighted in yellow

# Exercise Goals

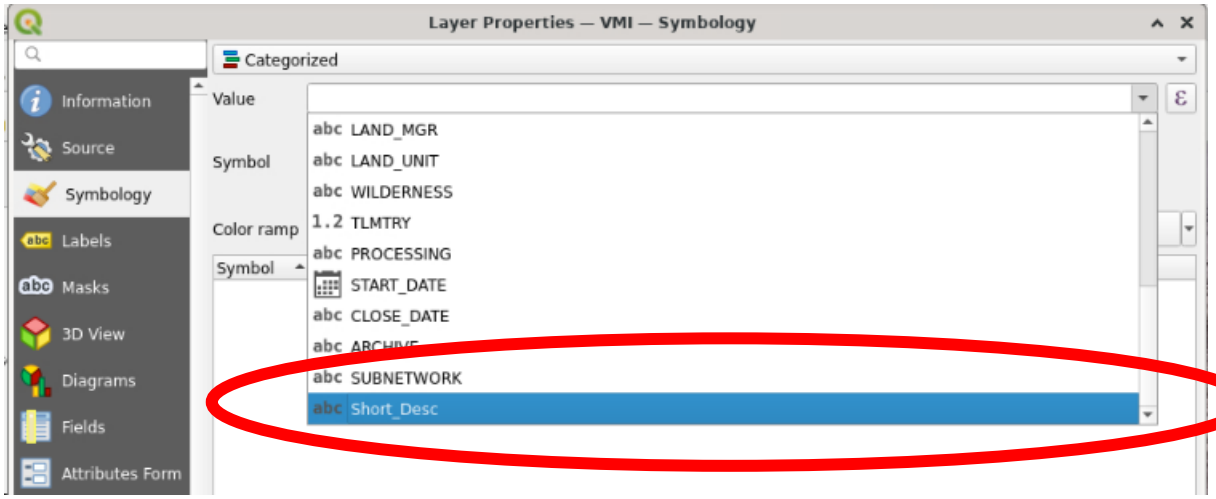
- View Instrument attribute table
- Select instruments by a drawing rectangle
- Select instruments by their attributes
- **Assign colors to instrument type**
- Make a map and add a legend to describe the meaning of colors



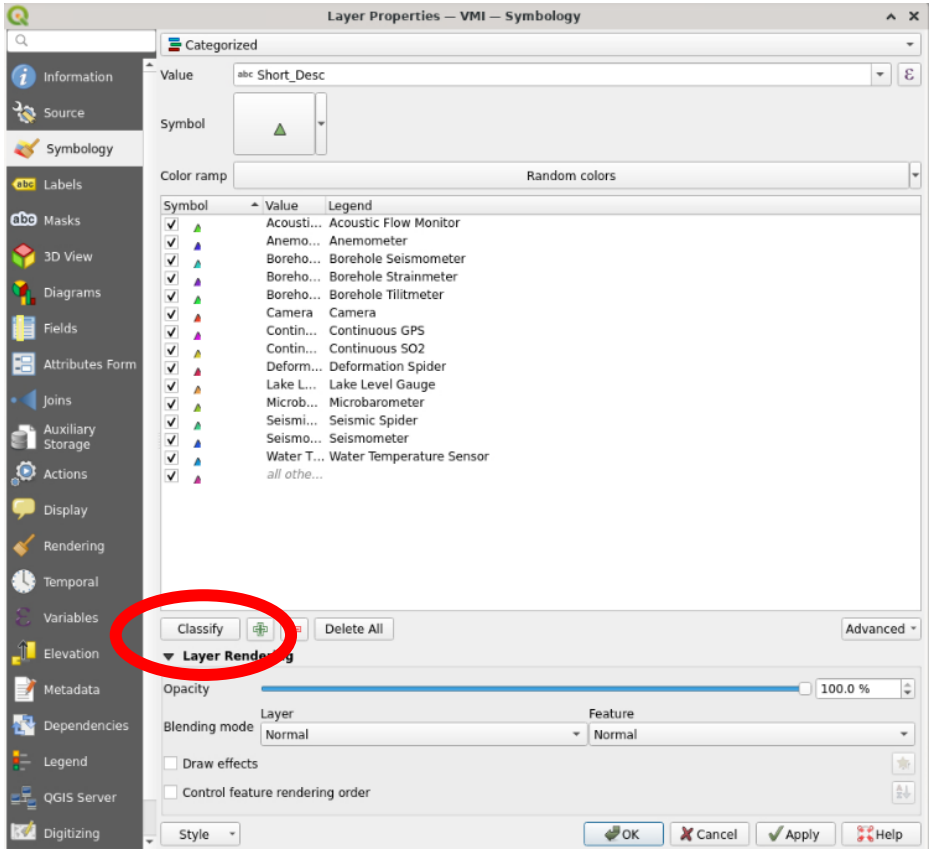

- We can use the attribute table to assign different symbols to the VMI layer so we can more easily tell what kind of stations these are
- Double click the VMI layer and open the Layer Properties

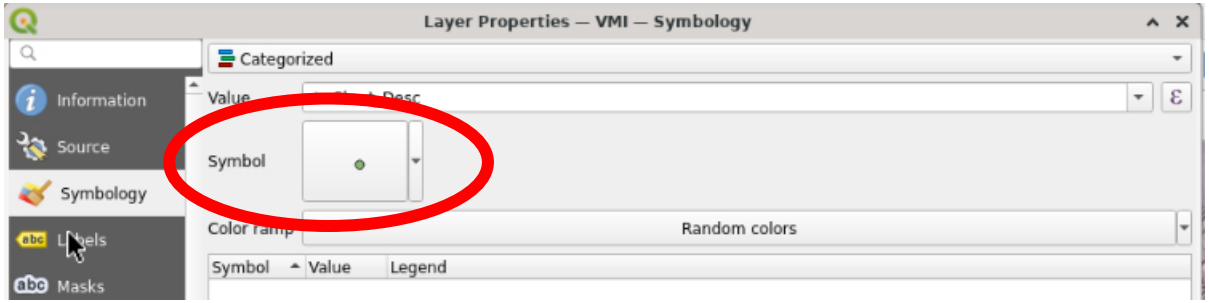


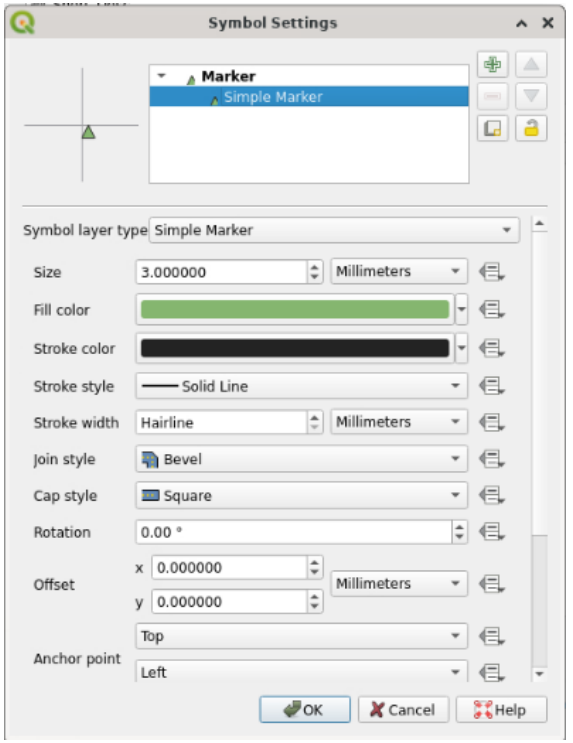

- Change from “Single symbol” to “Categorized”



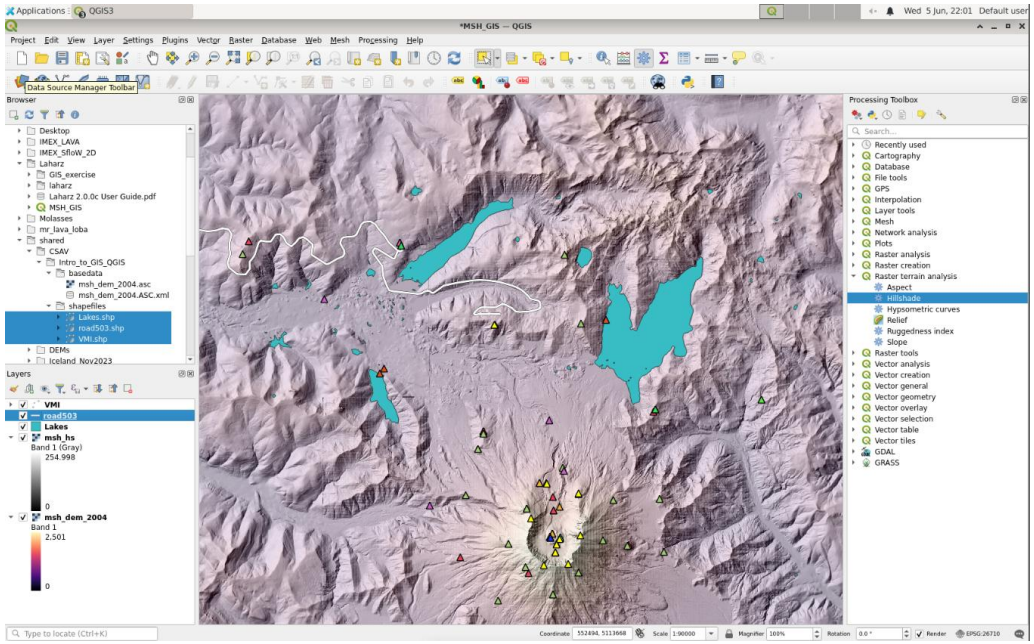

- Change the “Value” to the column you want to use for the display. We want to know what kind of stations these are, so choose “Short_desc”.
- Then click “classify” and all the different station types should appear





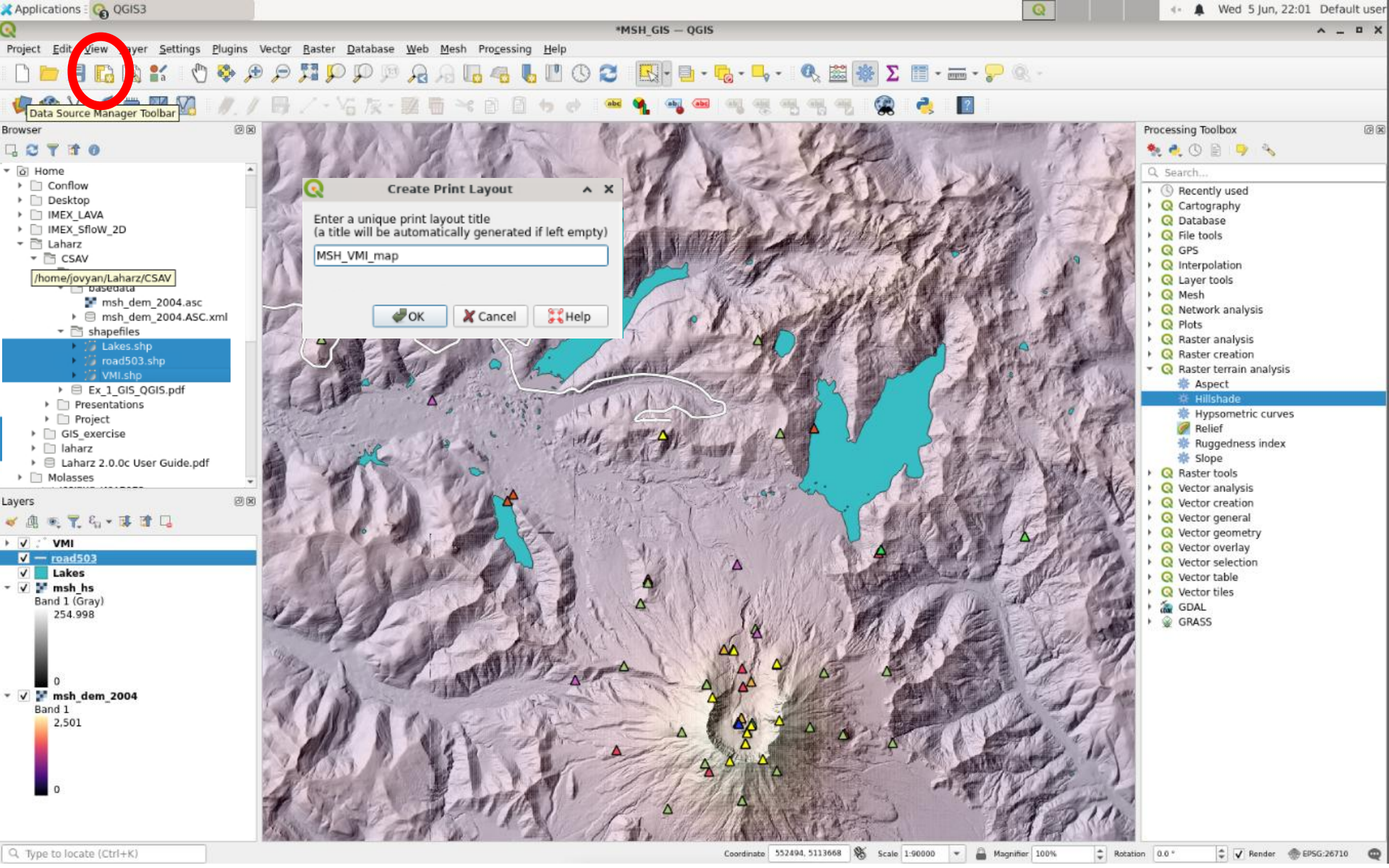




- Click the “Symbol” button to change the size and shape of the symbols. I chose 3 mm triangles
- Because station type is a categorical variable, choose “Random colors” for color.
- You can also change the colors for each symbol if you wanted
- Click “OK”
- Remember to save your project!

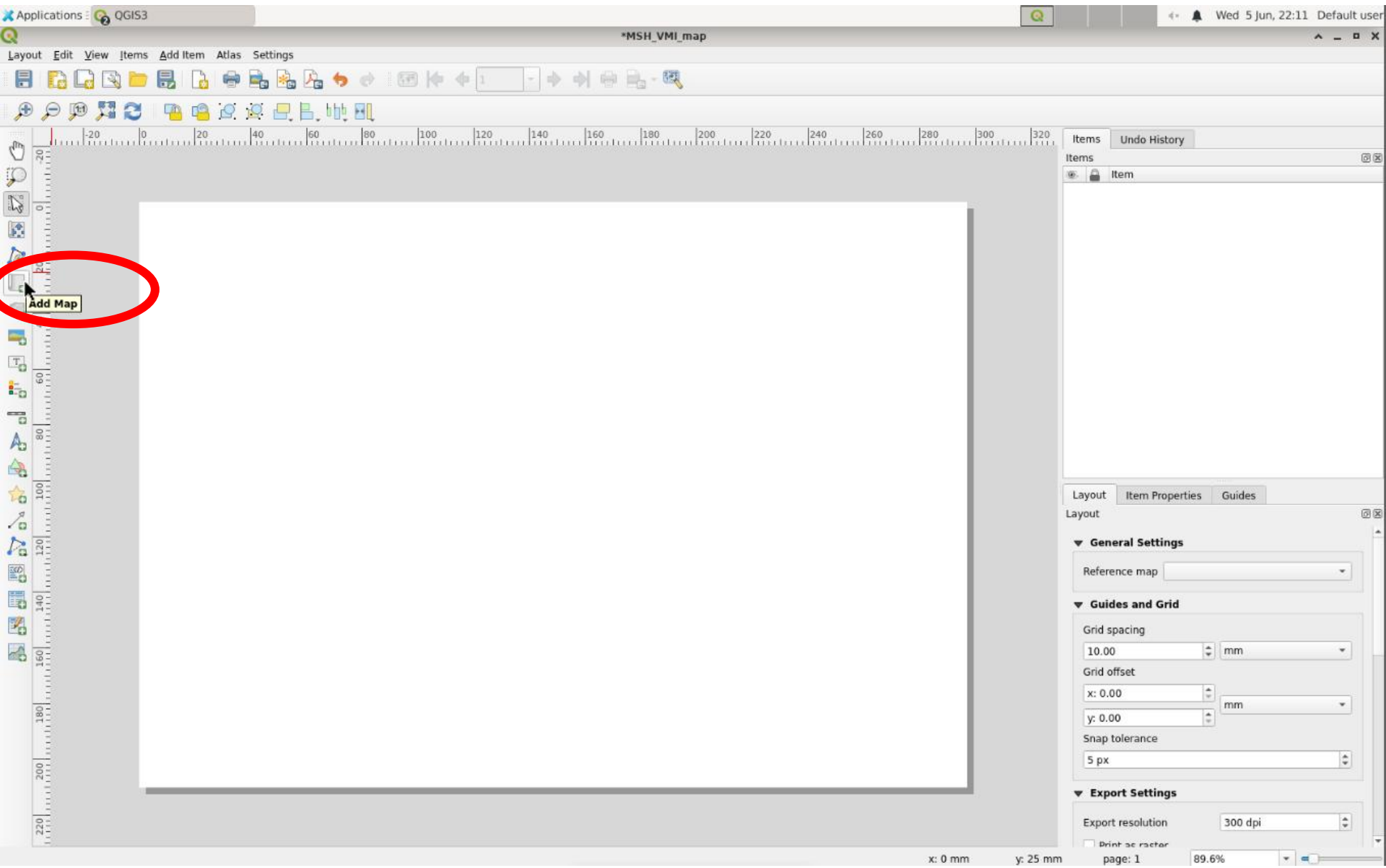


# Exercise Goals

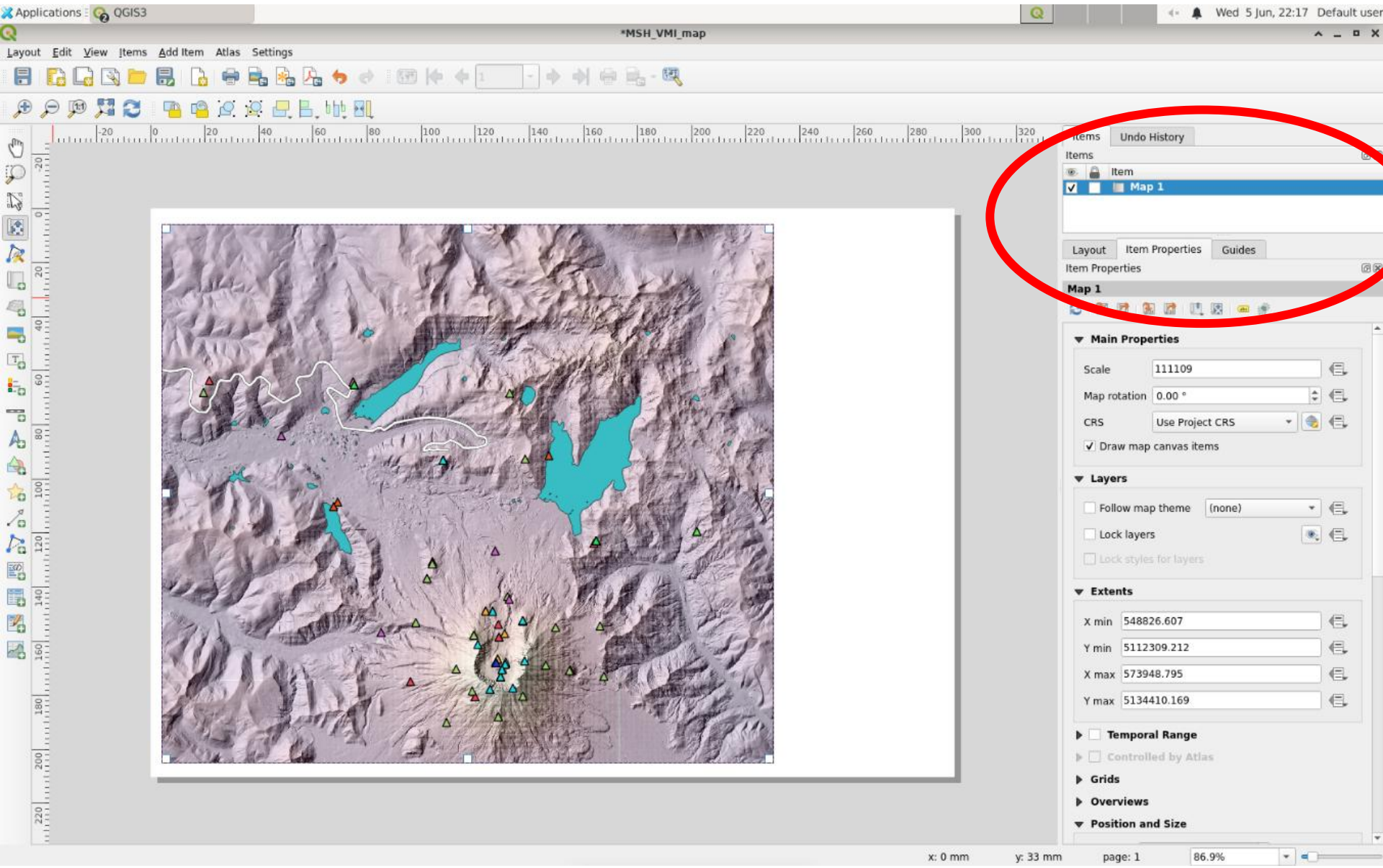
- View Instrument attribute table
- Select instruments by a drawing rectangle
- Select instruments by their attributes
- Assign colors to instrument type
- **Make a map and add a legend to describe the meaning of colors**



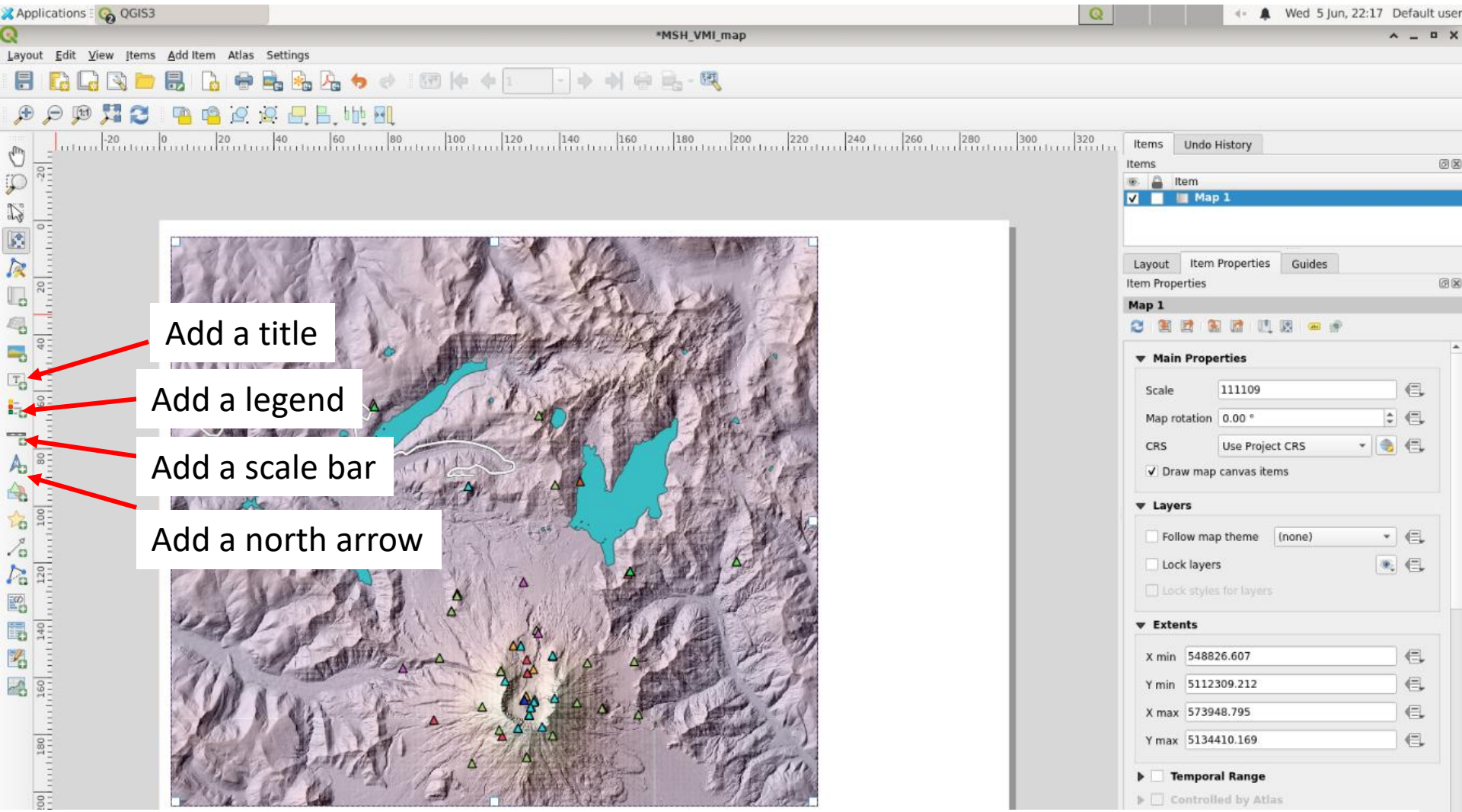

- Let's make a map!
- Click the "New Print Layout" button
- Enter a name for the new map. I chose "MSH_VMI_map"
- Click "OK"




- First you need to add your map to the layout
- Click the “Add map” button
- Then draw a rectangle on the blank page to add the map




- You can adjust the scale and extent and other properties by selecting the map layer and “Item properties”




- You can add a legend for the stations and other layers
- Add a scale bar and/pr coordinates
- Add a north arrow
- Play around with the options to make your map
- Save your final work!
- You can also export the map as an image